DRAW — User Manual

A pixel art editor written in QB64-PE — that exports your artwork as QB64 source code.

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Inspired by Dazzle Draw, Deluxe Paint, Grafx2, Inkscape, The GIMP, and yes... Photoshop

Table of Contents

About this Manual

This manual is the companion reference to the DRAW pixel art editor. It mirrors the structure of the upcoming 55-episode video tutorial series and walks you from "I just installed DRAW" through real-world pixel-art workflows, advanced layer techniques, and the quirky touches that only DRAW offers — like exporting your artwork as runnable QB64 source code.

You can read it cover-to-cover, jump to any chapter from the Table of Contents, or use it as a quick-reference once you know your way around. Every chapter ends with practical exercises and every keyboard shortcut you'll see in the manual is also listed in the appendix and the live, registry-generated `SHORTCUTS.md` (keyboard **and** mouse).

Who this is for

- 🎨 **Pixel artists** moving from DPaint / Aseprite / ProMotion who want a free open-source alternative.
- 🎮 **Game developers** who need fast sprite, tile, and palette workflows with deep export options.
- 💻 **QB64-PE / QBasic enthusiasts** who appreciate that the whole editor and its native export format is QB64 code.
- 🆕 **Newcomers to pixel art** who want a guided walkthrough rather than a feature dump.

Conventions used in this manual

Convention	Meaning
`Ctrl+S`	Press the keys together. On macOS, `Ctrl` is `⌘ Cmd` unless the manual says otherwise.
`B`	Press a single key (no modifier).
`[ ]`	Brackets around a key reference (often used for brush size keys).
Click / **Right-click** / **Middle-click**	Mouse buttons L / R / M.
Drag	Press and hold the left mouse button while moving.
🎯 *Goal*	What you'll be able to do after the section.
🎨 *Try it*	A short, practical exercise.
💡 *Tip*	Something useful but optional.
⚠️ *Gotcha*	Something that commonly trips people up.

How this manual is organized

```
docs/
├── MANUAL.md            ← this cover + master TOC
└── MANUAL/
    ├── 01-introduction.md       … 21-ai-generation.md
    ├── SCREENSHOTS.md   ← capture checklist for missing visuals
    └── images/          ← captured screenshots + placeholder.svg
```

If you are reading this on GitHub, every chapter link above will jump straight to the rendered chapter file. If you are reading offline in VS Code, open the Markdown preview (`Ctrl+Shift+V`) on this file and the entire manual is one click away.

Contributing & feedback

DRAW and this manual are open source. If you find a mistake, or want to add a workflow, please open an issue or pull request on the GitHub repository. Thank you for reading — now go make something pixelated.

— *grymmjack*

Ch. 01 🎬 Introduction & Setup

> **What you'll learn:** What DRAW is, why it exists, how to install or build it on Windows / Linux / macOS, and how to find your way around the interface in five minutes.

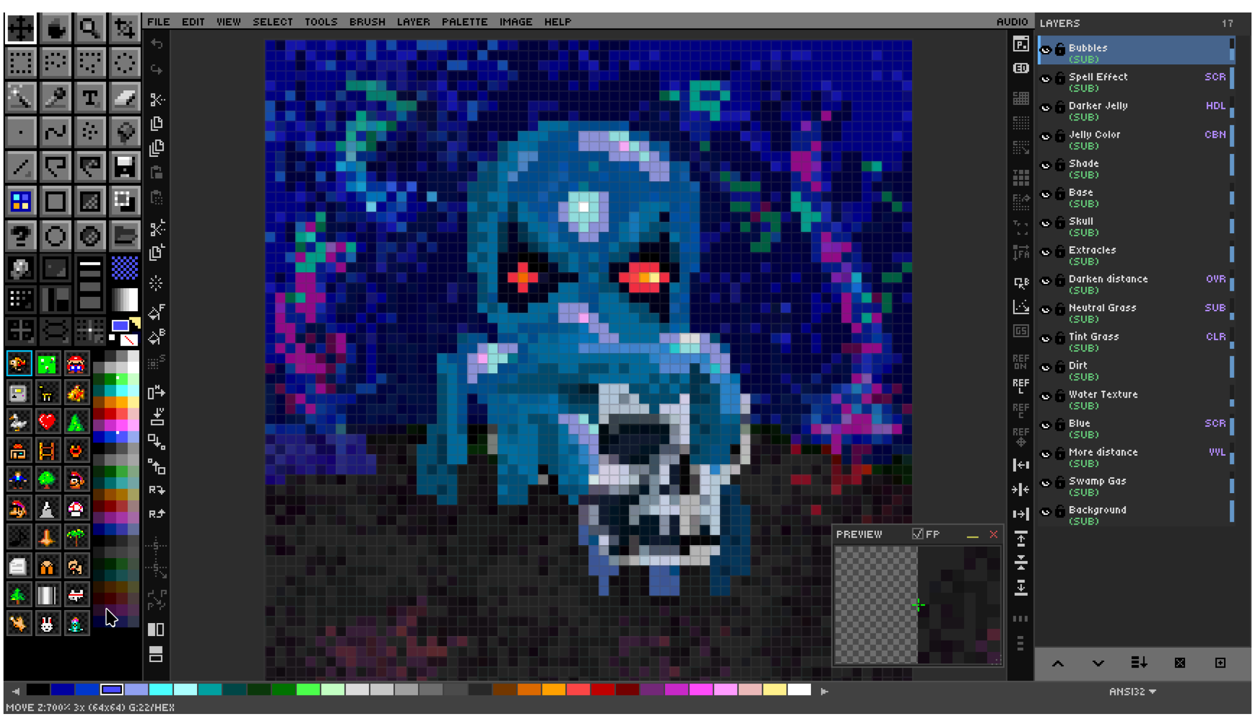
What Is DRAW?

> 🎯 **Goal:** Understand what DRAW is and why it exists.

DRAW is a pixel art editor written entirely in QB64-PE — a modern reimagining of Microsoft's QuickBASIC compiler. That alone would make it a curiosity. What makes it genuinely different is its signature party trick: **DRAW can export your pixel art as runnable QB64 source code**. Open the resulting `.bas` file in any text editor, hit compile, and your sprite is reborn as a self-contained BASIC program that draws itself pixel by pixel.

The project takes its inspiration from the legends of the genre — BR0DERBUND's **DazzleDraw** (on the Apple][), Electronic Arts' **Deluxe Paint** (DPaint DP2 enhanced for DOS), **Grafx2** by Sunset Design, Cosmigo's **ProMotion**, a little **Inkscape**, a little **GIMP**, and of course Adobe **Photoshop**, as well as the other DP spiritual heirs that followed including **DPaint.js**, etc. If you grew up with Amiga art, DOS demoscene tools, or the modern Aseprite, or LibreSprite, DRAW will feel familiar within seconds.

DRAW is **cross-platform** (Windows, Linux, macOS), **open source** (MIT licensed), and **dependency-free** at runtime — once it's built, you have a single executable plus its `ASSETS/` folder. It ships with **64 layers**, **19 blend modes**, a complete **text system** with rich formatting, **theming**, **audio**, **custom brushes**, and **symmetry drawing**. Its native `.draw` file format is a regular PNG file with a custom embedded chunk, so any image viewer can preview your art even though no other application can re-edit it. You don't need to save your file as .draw.png or .draw, as it recognizes draw files automatically.



Getting DRAW — Download, Build & Install

> 🎯 **Goal:** Get DRAW running on your machine.

There are two paths to a working install: **download a pre-built release** (recommended for almost everyone) or **build from source** (recommended if you want to hack on DRAW itself).

Option A — Pre-built release

Visit the DRAW Releases page and grab the archive that matches your platform. Each release ships as a self-contained zip containing the `DRAW` executable, the entire `ASSETS/` tree, and a default `DRAW.cfg`. There are no separate library installs, no runtime dependencies, and no internet connection required for normal operation.

Option B — Build from source

You will need the QB64-PE compiler (from github.com/QB64-Phoenix-Edition/QB64pe). You need v4.4+. Once it is installed and on your `PATH`:

```
git clone https://github.com/grymmjack/DRAW.git
cd DRAW
qb64pe -w -x -o DRAW.run DRAW.BAS
```

The repository also ships a `Makefile` with shortcuts (`make`, `make clean`, `make run`).

Platform-specific setup

- **Linux** — the `INSTALL/` folder contains a `.desktop` file and MIME-type registration for the `.draw` format so you can double-click `.draw` files in your file manager and have them open in DRAW.
- **Windows** — an installer registers the application in the Start Menu and associates `.draw` with the executable in the Registry.
- **macOS** — run `INSTALL/install-mac.command` once; it copies the app bundle into `/Applications` and sets up Launch Services associations.

First launch

On first run, DRAW probes your monitor with `_DESKTOPWIDTH` / `_DESKTOPHEIGHT` and chooses a sensible **display scale** (1× through 8×) and **toolbar scale** (1× through 4×). It writes a fresh `DRAW.cfg` next to the executable. That file is plain text — you can hand-edit it any time, and you can override the defaults entirely with `--config /path/to/your.cfg`.

Your First 5 Minutes — UI Tour

> 🎯 **Goal:** Navigate the interface confidently.

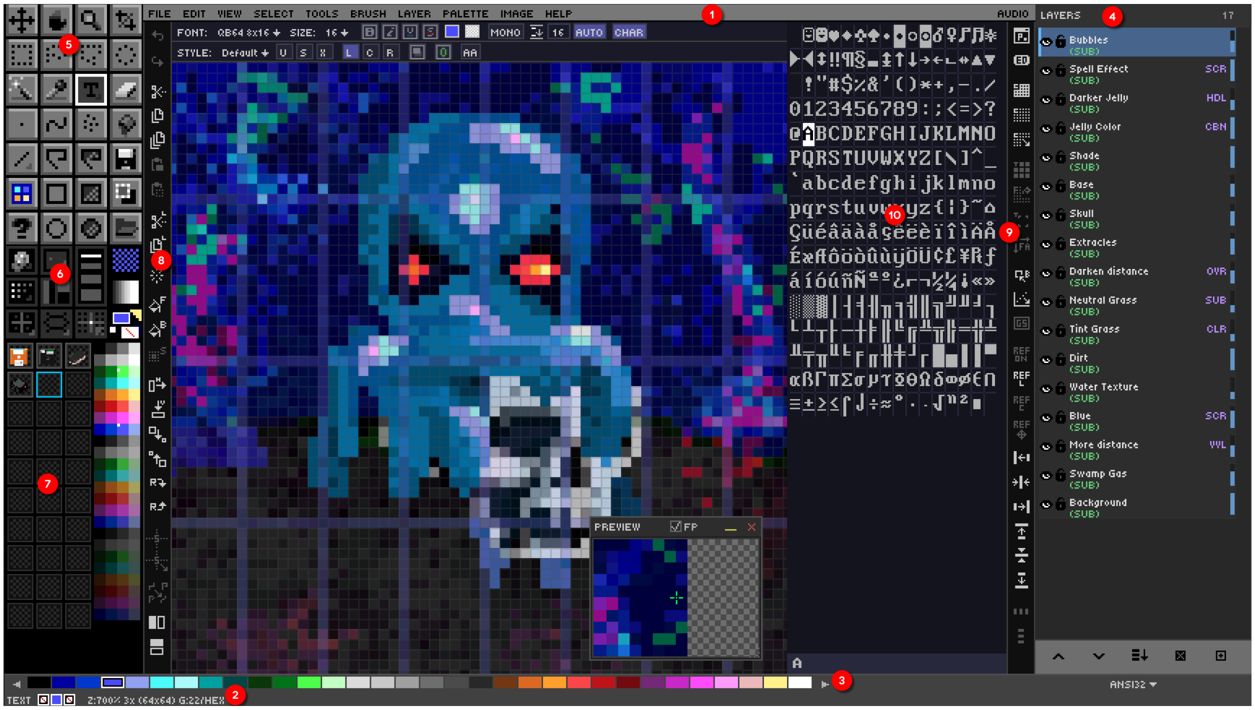
When DRAW launches you'll see six regions. From top to bottom and outside in:

Region	Purpose
Menu bar	11 menus from File through Audio.
Toolbar	A 4×7 grid of 28 tool icons on the left.
Canvas	The drawing surface, centered.
Layer panel	Stack of layers on the right.
Palette strip	The current palette across the bottom.
Status bar	Coordinates, zoom, current tool, grid state.

A second tier of **hidden panels** can be summoned on demand:

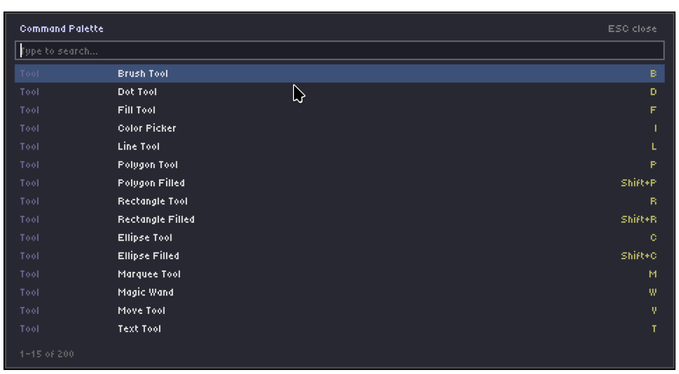
- **Organizer Widget** — brush presets and toggles.
- **Drawer Panel** — 30 reusable brush / pattern / gradient slots.
- **Edit Bar** (`F5`) — quick edit actions.
- **Advanced Bar** (`Shift+F5`) — 26+ toggles.
- **Preview Window** (`F4`) — magnifier or floating image.
- **Character Map** (`Ctrl+M`) — 16×16 glyph grid for text-mode work.

Every dockable panel can be flipped to the opposite side with **Ctrl+Shift+Click** on its title or icon area. `F11` toggles *all* UI for distraction-free drawing; `Ctrl+F11` keeps only the menu bar; `Tab` toggles the toolbar alone.



1. Menu Bar
2. Status Bar
3. Palette Strip
4. Layer Panel
5. Tool Box
6. Organizer
7. Drawer Bins
8. Edit Bar
9. Advanced Bar
10. Character Map

The single most important keystroke to remember on day one is `?` — that opens the **Command Palette**, a fuzzy-searchable list of every command in the app. If you can't remember a shortcut, type a few letters and hit Enter.



➡️ Next: Chapter 2 — Core Drawing Fundamentals

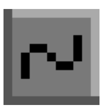
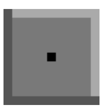
Ch. 02 🖌️ Core Drawing Fundamentals

> **What you'll learn:** Free-hand brushwork, geometric shapes, polygons, the flood-fill tool, the spray can, and the eraser — the seven tools that account for 90% of pixel art.

Brush, Dot & Freehand Drawing

> 🎯 **Goal:** Master basic freehand pixel drawing.

The two beginner tools you'll meet first are the **Brush** (`B`) for freehand strokes and the **Dot** (`D`) for single-pixel placement. Both share the same brush size and shape state.

Tool	Icon	Key	Behaviour
Brush		`B`	Drag to paint a continuous stroke.
Dot		`D`	Click to stamp one pixel (or one brush footprint).

Brush size runs from 1 to 50 pixels and is adjusted with `[` (smaller) and `]` (larger). The **brush shape** toggles between circle and square with `\`. To preview the current footprint and color before committing, hold the backtick key (`` ` ``).

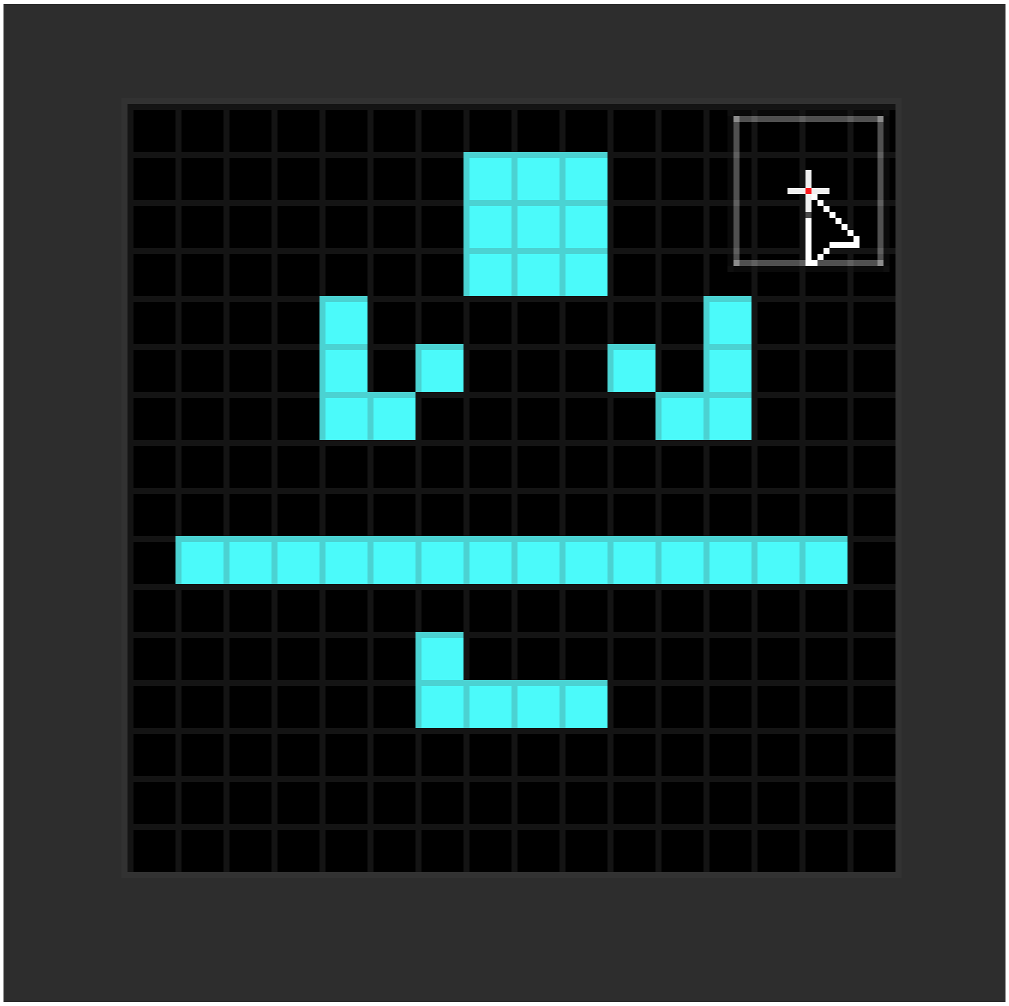
DRAW follows the universal "left = foreground, right = background" convention. **Left-click paints with the FG color**, **right-click paints with the BG color**. Holding `Shift` while drawing constrains the stroke to a perfectly horizontal or vertical axis. Holding `Shift` and right-clicking draws a **connecting line** from the previous click to the current position — invaluable for stitching together long straight lines without ever touching the Line tool.

If you have a steady hand but find that single-pixel stairsteps creep into your strokes, enable **Pixel Perfect mode** with `F6`. DRAW will retroactively remove L-shaped corners as you draw, leaving cleaner outlines.

The Organizer widget on the left contains **four brush size presets** (1, 3, 5, 9 pixels for sprite work).

🎨 **Try it — your first 16×16 sprite**

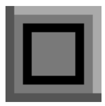
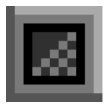
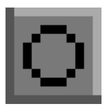
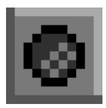
1. `Ctrl+N` for a new canvas at 16×16 pixels.
2. Press `D` for the Dot tool, set size to 1 with `[`.
3. Sketch out the silhouette of a simple character.
4. Switch to `B` (Brush), increase to size 2, and fill the larger regions.
5. Save as `.draw` so you can come back later.



Lines, Rectangles & Ellipses

> 🎯 **Goal:** Draw clean geometric shapes.

When freehand stops being precise enough you reach for the geometric trio. All three respect the current brush size, color, and symmetry settings.

Tool	Icon	Outlined	Filled
Line		`L`	—
Rectangle	 	`R`	`Shift+R`
Ellipse / Circle	 	`C`	`Shift+C`

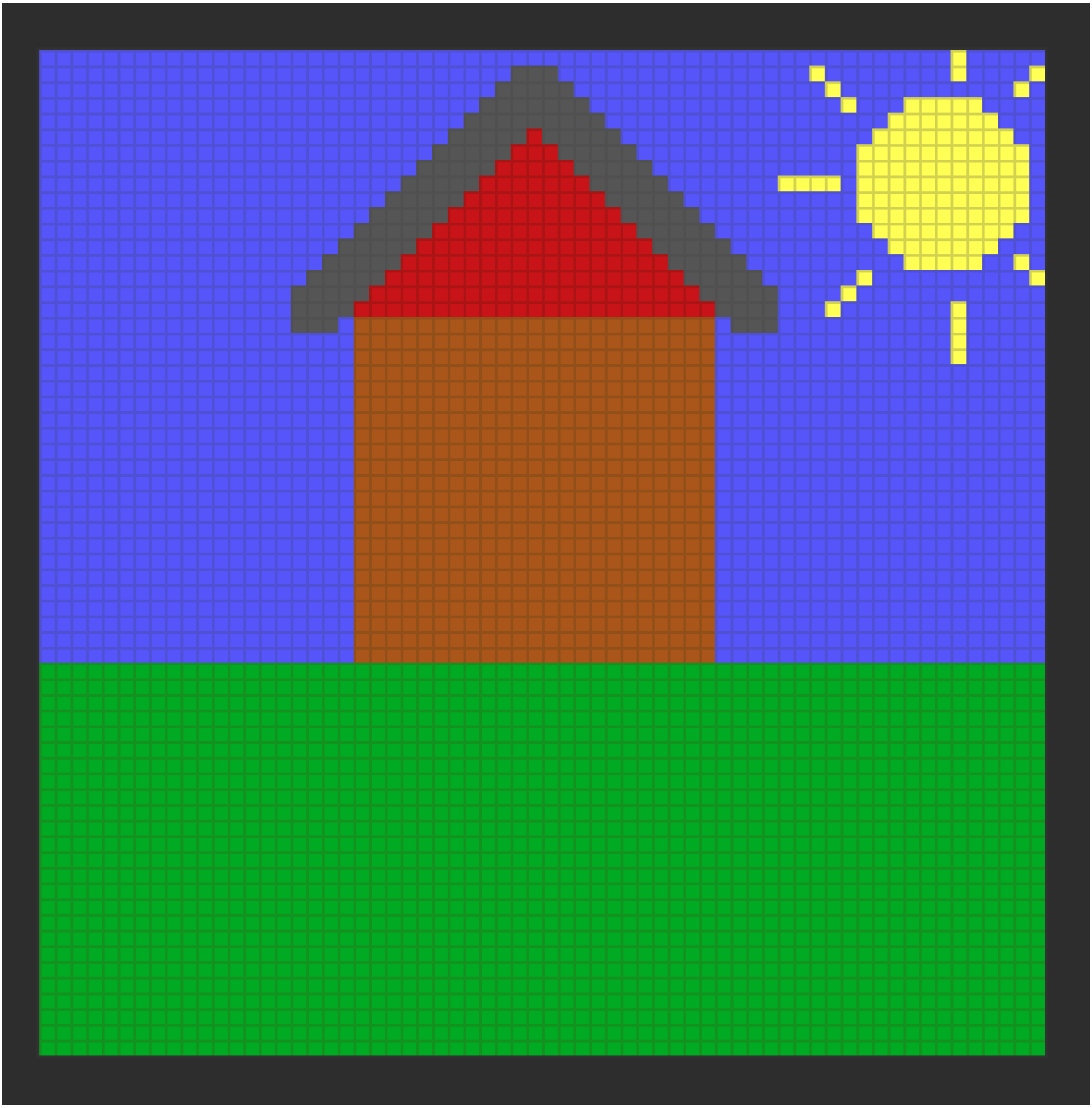
Tip: While dragging any shape, hold SPACE to move the entire shape with the mouse instead of resizing it. Release SPACE to resume resizing from the new anchor. This works for all geometric and Smart Shape tools, and honors grid snap.

The shared modifier vocabulary across all three:

- **Shift** — constrain (line: H/V; rect: square; ellipse: circle).
- **Ctrl** — perfect aspect (forces square / circle even after you start dragging).
- **Ctrl+Shift while drawing the rect or ellipse** — anchor from the *center* instead of the corner.
- **Ctrl+Shift while drawing a line** — angle snap to 15°/30°/45°/90° increments.

🎨 **Try it — house in 30 seconds**

1. Filled rectangle for the body. `Shift+R`
2. Two lines forming the roof triangle. `L`
3. Filled circle (the sun). `Shift+C`
4. A few short lines for the sun's rays. `L`
5. Filled rectangle for the ground. `Shift+R`



Polygons & the Fill Tool

> 🎯 **Goal:** Draw complex shapes and fill regions.

The **Polygon** tool comes in two flavours: outlined (`P`) and filled (`Shift+P`). Each click adds a vertex to the in-progress polygon; press `Enter` to close and commit. As with the line tool, **Ctrl+Shift** snaps the segment angle.

The **Flood Fill** tool (`F` , ) pours the FG color into every pixel contiguous with the click point that shares its starting color. By default it samples only the active layer; hold `Shift` to sample from the **merged visible composite** (useful when you have outlines and fills on separate layers).

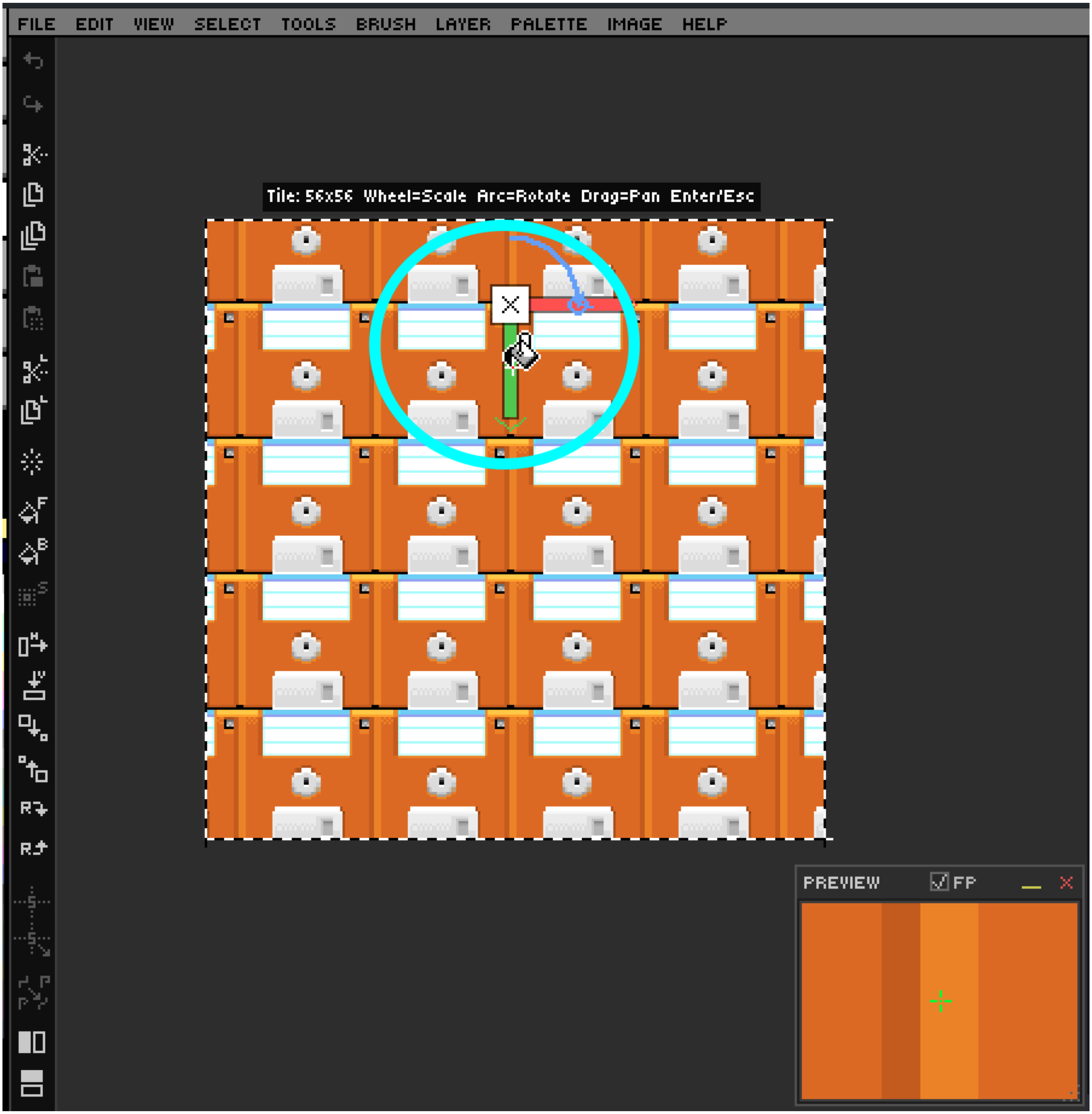
DRAW's flood fill is unusual in two ways:

1. It supports **custom brushes as a tiled fill** — if you have a brush captured (Chapter 9), the fill is rendered as a seamless tiling of that brush rather than a flat color.
2. It also honours the **paint mode** — set Pattern or Gradient mode (Chapter 9) and your fill becomes a dithered gradient or tiled pattern instead of a solid color.

Press `F8` to enable **Fill Adjustment overlay**. Now when you fill using pattern mode or gradient mode, or with custom brushes in color mode (loaded from the bins), drag the canvas to reposition the tile origin, mouse-wheel for uniform scale, drag the L-handle for independent X/Y scaling, and drag the rotation handle (small arc) to rotate the tile. `Enter` applies, `Esc` cancels.

🎨 **Try it — tileable pattern fill**

1. Draw a small 8×8 motif.
2. Capture it as a custom brush (see Chapter 9).
3. On a fresh layer, flood-fill a large rectangle with the brush active.
4. Press `F8` and experiment with scale, rotation, and offset.



Spray Tool & Eraser

> 🎯 **Goal:** Use spray-can effects and clean up mistakes.

The **Spray** tool (`K` ,) emits randomized dots within a circular nozzle. The nozzle radius **doubles for every brush-size step**, and density scales with radius — small brush = pinpoint mist, large brush = thick coverage. Spray respects custom brushes too: every "drop" stamps the brush instead of a single pixel, which is wonderful for foliage, dust, and confetti.

The **Eraser** (`E` ,) paints fully transparent pixels — it does not paint with the BG color, it removes alpha. Two non-obvious tricks make it indispensable:

- **Hold `E` to temporarily eraser anything** while a different tool is active. Release `E` to return to your previous tool. This is the fastest way to nudge an outline or clean a stray pixel.
- **Smart Erase** (`Shift` while erasing) operates on **all visible layers at once**, with per-layer history tracking so each layer's undo step is independent.

The eraser uses your current brush size, shape, and even custom-brush stamp. The status bar shows `FG:TRN` to remind you transparent painting is active.

> 💡 **Tip:** Combine the eraser with a layer's **Opacity Lock** (Chapter 4) to clean up overpaint while preserving the layer's silhouette.

Smart Shapes & Advanced Tools

> 🎯 **Goal:** Draw parametric shapes, 3D dice, and advanced curves.

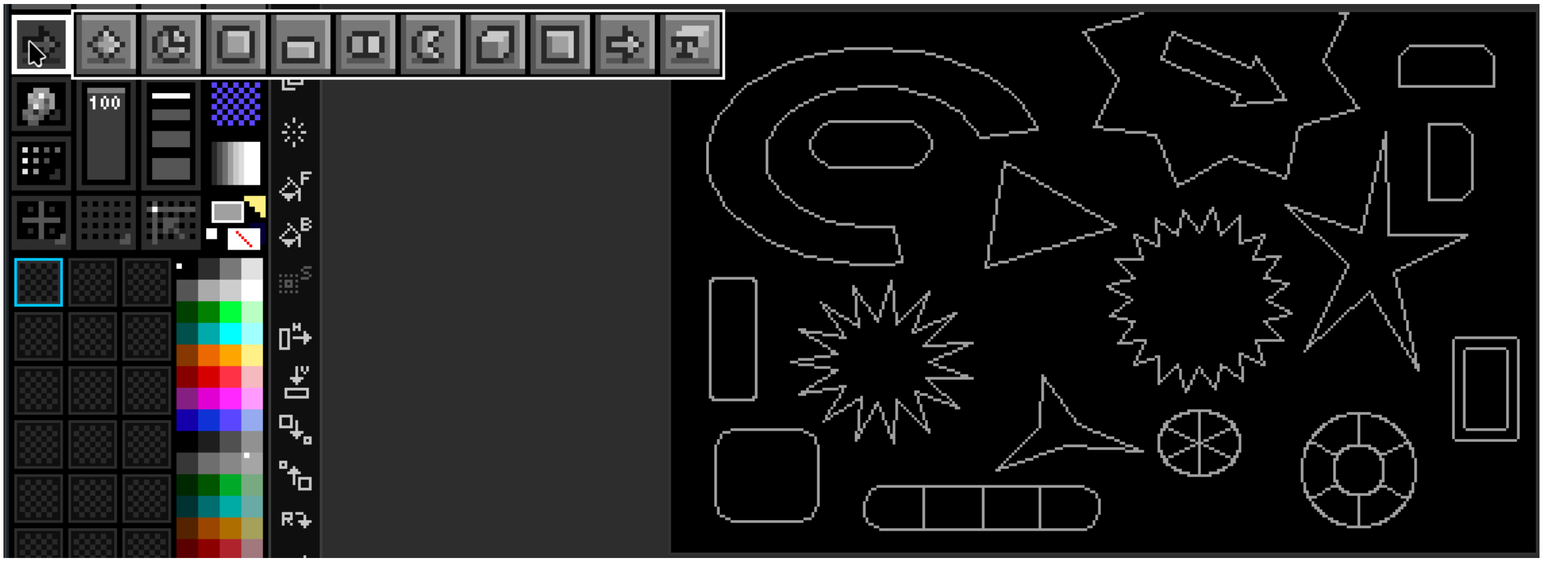
Smart Shapes Tool Group

The **Smart Shapes** tool group is accessed from the toolbar. **Long-press or right-click** the Smart Shapes button to open a flyout menu and pick a sub-tool. Each shape has live drag-time arrow-key tweaks and unique modifiers.

Tip: While dragging any Smart Shape, hold SPACE to move the entire shape with the mouse instead of resizing it. Release SPACE to resume resizing from the new anchor. Grid snap is honored.

Sub-tools:

Smart Shapes — Detailed Controls & Behaviors



The idea of smart shapes is that you begin drawing, then use keys to modify the shape as you create it. All smart shapes snap to the grid and honor grid/symmetry settings, just like regular rectangles and ellipses.

Below are the controls and behaviors for each smart shape:

Polygon

> Minimum sides: 3, Maximum: 30

Draws a regular polygon. While dragging, use:

- **Up arrow**: Add edges
- **Down arrow**: Remove edges
- **Left arrow**: Decrease point/star depth (makes a star/burst)
- **Right arrow**: Increase point/star depth

Pie / Donut

> Create pie charts or donut shapes with equidistant or custom slices. Left/right arrows add a hole (donut mode). Minimum slices: 0, Maximum: 30. Max hole radius: 90% of shape size.

- **Up arrow**: Add segments
- **Down arrow**: Remove segments
- **Left arrow**: Decrease hole size
- **Right arrow**: Increase hole size

Rounded Rectangle

> Rectangle with rounded corners. Corner radius can be 1 (almost square) up to 30 (fully rounded), but never so large that corners intersect.

- **Up arrow**: Increase corner radius
- **Down arrow**: Decrease corner radius

Tab

> Rounded rectangle with one flat edge. Flat edge can be cycled to any side (top, bottom, left, right).

- **Up arrow**: Round top corners more
- **Down arrow**: Round top corners less
- **Left arrow**: Cycle flat edge left
- **Right arrow**: Cycle flat edge right

Pill

> Capsule/oval with dividers for options (like a segmented control). Horizontal by default.

- **Up arrow**: Increase roundness
- **Down arrow**: Decrease roundness
- **Left arrow**: Decrease segments
- **Right arrow**: Increase segments

Pac-Man

> Pie chart with a single slice cut out (the mouth). Mouth can be opened/closed. Inner hole for dial/knob effects.

- **Up arrow**: Increase mouth size
- **Down arrow**: Decrease mouth size
- **Left arrow**: Decrease inner hole
- **Right arrow**: Increase inner hole

3D Cube / 3D Polygon (Dice)

> Draws a 3D cube or polyhedral dice wireframe. Press number keys while dragging for dice types: 4 = D4, 6 = D6, 8 = D8, 0 = D10, 1 = D12, 2 = D20, 3 = D30.

- **Up arrow**: Increase Z-depth
- **Down arrow**: Decrease Z-depth
- **Left arrow**: Rotate left
- **Right arrow**: Rotate right
- **Mouse wheel up**: Rotate up
- **Mouse wheel down**: Decrease Z-depth (angle)

Bevel Rect

> Rectangle with beveled (angled) edges. Bevel and border size adjustable. Can switch between inner/outer bevel.

- **Up arrow**: Increase bevel size
- **Down arrow**: Decrease bevel size
- **Left arrow**: Decrease border size
- **Right arrow**: Increase border size
- **I**: Switch to inner bevel
- **O**: Switch to outer bevel
- **Mouse wheel up**: Increase Z-depth (angle)
- **Mouse wheel down**: Decrease Z-depth (angle)

Arrow

> Arrow with adjustable stem and head. Head can be made concave/curved.

- **Up arrow**: Make stem fatter
- **Down arrow**: Make stem skinnier
- **Left arrow**: Make arrow head shorter
- **Right arrow**: Make arrow head longer
- **Mouse wheel up**: Increase head concavity
- **Mouse wheel down**: Decrease head concavity

- **Right-click drag** — solid mode (BG color fill, FG color wireframe)
- Dice type: `4` / `6` / `8` / `0` / `1` / `2` for D4/D6/D8/D10/D12/D20

Bezier Curve Tool (`Q`)

- `Escape` — cancel

Line Tool — End Caps

While dragging the Line tool, press `s` / `e` to cycle the start / end cap shape (none → arrow → diamond → circle → square → ...). End caps render in the same FG color as the line.

➡️ Next: Chapter 3 — Color & Palette Mastery

Ch. 03 🎨 Color & Palette Mastery

> **What you'll learn:** How DRAW thinks about color, the FG/BG/swap dance, the Color Picker and live Color Mixer, the 56 bundled palettes, and the palette-ops mode that lets you remap entire compositions in seconds.

Color Basics — FG, BG & Palette Strip

> 🎯 **Goal:** Select, swap, and manage colors.

The palette strip across the bottom of the screen is the fastest way to choose colors. **Left-click a swatch** to set the foreground (FG) color; **right-click a swatch** to set the background (BG) color. The active FG and BG swatches are echoed in the status bar.

Action	Key / mouse
Swap FG and BG	`X`
Reset to white FG / black BG	*Palette menu → Default Colors*
Set BG to transparent	`Shift+Delete`
Scroll the palette strip	Mouse wheel over strip
Fast-scroll 32 colors at a time	`Shift` + Mouse wheel
Temporary FG eyedropper (any tool)	`Alt` +Left-click on canvas
Temporary BG eyedropper (any tool)	`Alt` +Right-click on canvas

Paint opacity — the keys `1` through `9` set strokes to 10–90% opacity, and `0` returns to fully opaque 100%. DRAW uses **per-stroke compositing**, which means a 50% stroke that overlaps itself does not get *more* opaque the way Photoshop's brush would; the stroke is rendered once at the end as a single 50% pass. This makes for predictable, pixel-art-friendly results.

The Opacity Slider

Use the opacity slider in the organizer of the toolbox to set whatever opacity level you wish from 1% to 100%.

Action	Key / mouse
Set Opacity from 10% to 100%	`1-9` for 10% to 90%, or `0` for 100% opacity
Mouse Wheel Up	Increase opacity by 1%
Mouse Wheel Down	Decrease opacity by 1%
`Shift` + Mousewheel	Increase or decrease opacity by 10%

> Opacity slider will be outlined in a bright color if you set opacity to `⩽` the percentage according to `DRAW.cfg` setting: `OPACITY_WARN_THRESHOLD`

Color Picker, Color Mixer & Custom RGB

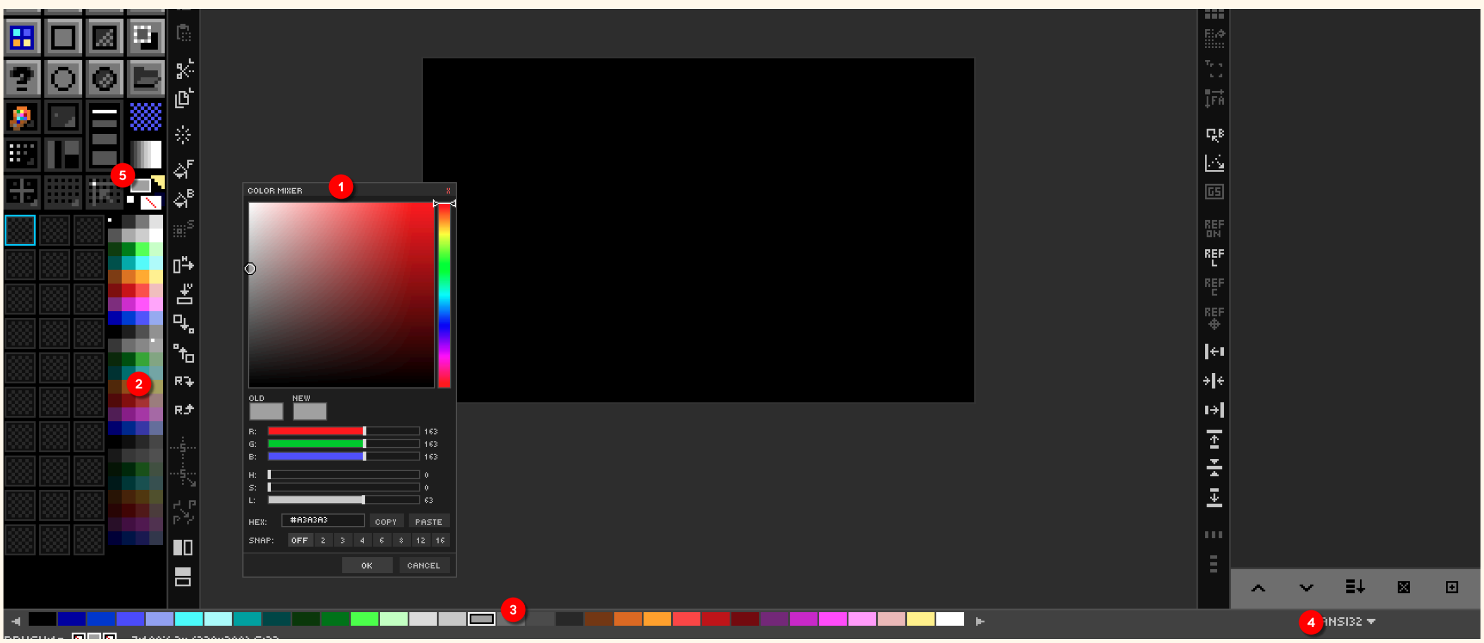
> 🎯 **Goal:** Use the full RGB color picker and the live Color Mixer.

Click either the FG or BG swatch in the status bar to open the **Color Picker**. It supports true 24-bit RGB selection plus a hex input field. The Picker tool (`I`) on the canvas pairs an eyedropper with a **loupe overlay** that magnifies the area under the cursor and prints the RGB and hex values for the pixel you'd sample.

The **Color Mixer panel** is a floating, persistent alternative to the modal picker. Open it from `View → Color Mixer` (or via the Command Palette). It exposes:

- RGB sliders — adjust Red, Green, Blue independently.
- HSV sliders — adjust Hue, Saturation, Value.
- A hex input field — paste any hex value directly.
- FG and BG swatches — click to apply the mixed color.

Because the mixer is non-modal, you can keep it open while drawing and tweak colors live. Its visibility is persisted in `DRAW.cfg`.

> DRAW's various color widgets and doo-dads
>
> 
>
>
> 1. Color Mixer — `View → Color Mixer`
> 2. Mini Palette — Always visible as part of Tool Box
> 3. Color Strip — Can be hidden, shows (4) Palette
> 4. Palette Picker — Choose a palette for editing
> 5. Active Foreground (FG) and Background (BG) colors

Palette Management — 56 Built-in Palettes

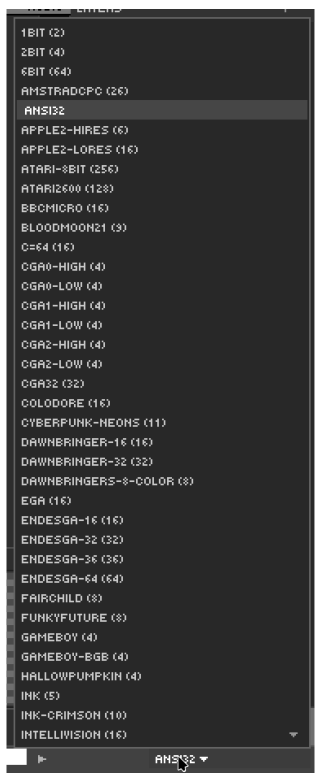
> 🎯 **Goal:** Switch, browse, and manage palettes.

DRAW ships with **56 palettes** in GIMP's `.gpl` format. They live under `ASSETS/PALETTES/` and include classics — **NES**, **PICO-8**, **Commodore 64**, **Game Boy**, **Endesga 32/64**, **DawnBringer 16/32**, **AAP-64**, **Sweetie 16**, **Resurrect 64**, **CGA**, **EGA**, **VGA**, **Amiga**, **MSX** — plus 40 more.

Click the palette name above the strip to open the dropdown. **Pressing a letter** while the dropdown is open jumps to the next palette starting with that letter — handy when you know the name but not the position.

Palette workflows DRAW supports natively:

- **Download from Lospec** — DRAW can fetch palettes from Lospec's online database directly.
- **Create palette from existing image** — distill the unique colors of any open image into a new palette.
- **Import / Export `.gpl`** — interchange with GIMP, Aseprite, Krita.
- **Remap existing artwork** — recolor a finished piece into a different palette while preserving structure.



Palette Ops — Edit Colors Directly

> 🎯 **Goal:** Modify palette colors and remap on canvas.

Palette Ops mode is one of DRAW's signature features. Toggle it from the Organizer panel. Once active, the palette strip becomes editable *and the canvas is remapped live as you change the palette*.

Gesture on a palette swatch	Effect
Double-click	Open color picker; new color is **substituted on the canvas** wherever the old one appeared.
Right-click	Place a marker / indicator on the swatch (visual bookmarking).
Middle-click	Delete the color and remap matching pixels to the nearest remaining color.
`Shift` +Middle-click	Insert a transparent (alpha 0) entry at this index.
Drag onto another swatch	Rearrange palette order.
Left-click	Magic-wand select all matching pixels on the active layer.

When you first enter Palette Ops, DRAW automatically creates a **`[DOCUMENT]` palette** that snapshots the current state. This means experimentation is safe — you can hop back to the original palette at any time without losing your remapping.

> 🎨 **Try it — colorway exploration**
>
> 1. Open a finished sprite.
> 2. Toggle Palette Ops.
> 3. Double-click each palette swatch in turn and shift the hue.
> 4. Compare colorways. When one feels right, exit Palette Ops to bake it in.

➡️ Next: Chapter 4 — Layer System Deep Dive

Ch. 04 📚 Layer System Deep Dive

> **What you'll learn:** How to use DRAW's 64-layer system, opacity and visibility, the 19 blend modes, layer groups, multi-layer selection, alignment, and the unique **symbol layer** system for repeated artwork.

Layer Basics — Create, Delete & Reorder

> 🎯 **Goal:** Work with multiple layers.

DRAW supports up to **64 layers** per canvas. The Layer Panel on the right lists every layer top-to-bottom in render order — top of the list is on top of the canvas.

Action	Shortcut
New Layer	`Ctrl+Shift+N`
Duplicate Layer	`Ctrl+Shift+D`
Delete Layer	`Ctrl+Shift+Delete`
Move Up / Down in stack	`Ctrl+PgUp` / `Ctrl+PgDn`
Move to top / bottom	`Ctrl+Shift+]` / `Ctrl+Shift+[`
Rename	Right-click → Rename

You can also **drag a layer row** with the left mouse button to reorder. **Right-click** any layer for the full context menu (rename, duplicate, delete, group, blend mode, etc.).

> 🎨 **Try it — multi-layer sprite**
>
> 1. Layer 1: scene/background.
> 2. Layer 2: character outline.
> 3. Layer 3: character base colors (with Opacity Lock — see below).
> 4. Layer 4: highlights and effects.

Opacity, Visibility & Opacity Lock

> 🎯 **Goal:** Control transparency and protect existing pixels.

Each layer row has three persistent controls:

- **Eye icon** — toggle visibility. **Alt+Click the eye** for **Solo mode**: hide every other layer. Drag horizontally across multiple eye icons to sweep the visibility state.
- **Opacity bar** — drag, or hover and use the mouse wheel, for 0–100%.
- **Lock icon** — **Opacity Lock**. When enabled, drawing tools can only modify pixels that are *already opaque*. Perfect for recoloring within a finalized silhouette without bleeding outside.

> 🎨 **Try it — opacity-locked recolor**
>
> 1. Draw a character on one layer (transparent background).
> 2. Toggle the lock icon on that layer.
> 3. Take a giant brush and slosh new color over the whole canvas. Notice it only paints inside the silhouette.

Blend Modes — All 19 Explained

> 🎯 **Goal:** Use blend modes creatively.

A layer's blend mode determines how its pixels combine with the composite below it. DRAW implements all the modes you would expect from a modern raster editor:

Family	Modes
Basic	Normal · Multiply · Screen · Overlay
Math	Add (Linear Dodge) · Subtract · Difference
Comparison	Darken · Lighten
Dodge / Burn	Color Dodge · Color Burn
Light	Hard Light · Soft Light · Vivid Light · Linear Light · Pin Light
Color / Difference variants	Exclusion · Color · Luminosity
Group only	Pass Through (Chapter 4 / EP15)

Common pairings:

- **Multiply** — for shadows and ambient occlusion. Multiplying any color by a darker color produces a richer darker color.
- **Screen / Add** — for highlights, glow, fire and lights. Screen is the gentler companion; Add can blow out very quickly.
- **Overlay** — for contrast and saturation boost passes.
- **Color** — apply hue and saturation while preserving the underlying luminosity. The classic photograph-tinting trick.
- **Luminosity** — the inverse: apply brightness while preserving the underlying color.

To cycle a layer's blend mode without opening the dropdown, `Shift` **+Right-click** on the layer row.

> 🎨 **Try it — three-layer lighting study**
>
> 1. Base sprite on layer 1, Normal mode.
> 2. Layer 2: Multiply mode, paint cool desaturated tones for shadows.
> 3. Layer 3: Screen (or Add) mode, paint warm highlights.

Layer Groups, Symbols & Advanced Operations

> 🎯 **Goal:** Organize complex artwork.

Groups

Action	Shortcut
Create empty group	`Ctrl+G`
Group selected layers	`Ctrl+Shift+G`
Ungroup	`Ctrl+Shift+U`

Groups can be nested arbitrarily and use a special **Pass Through** blend mode that lets each child blend naturally with what's below the group.

Merge operations

- **Merge Down** — `Ctrl+Alt+E`
- **Merge Visible** — `Ctrl+Alt+Shift+E`
- **Merge Selected** — fold a multi-selection into one layer.
- **Merge Group to Single Layer** — flatten a group while preserving its visual result.

Multi-layer selection

`Ctrl+Click` toggles individual selection. `Shift+Click` also works like `Ctrl` because Mac users need that. Once multiple layers are selected, every relevant operation — clear, fill, flip, scale, rotate — runs against the whole selection. There is also `Select All in Group`.

Alignment & distribution

DRAW provides nine alignment commands (Left/Center/Right × Top/Middle/Bottom) and two distribution commands (horizontal, vertical). Combined with multi-layer selection, this is the fastest way to lay out drawn objects in pleasing ways.

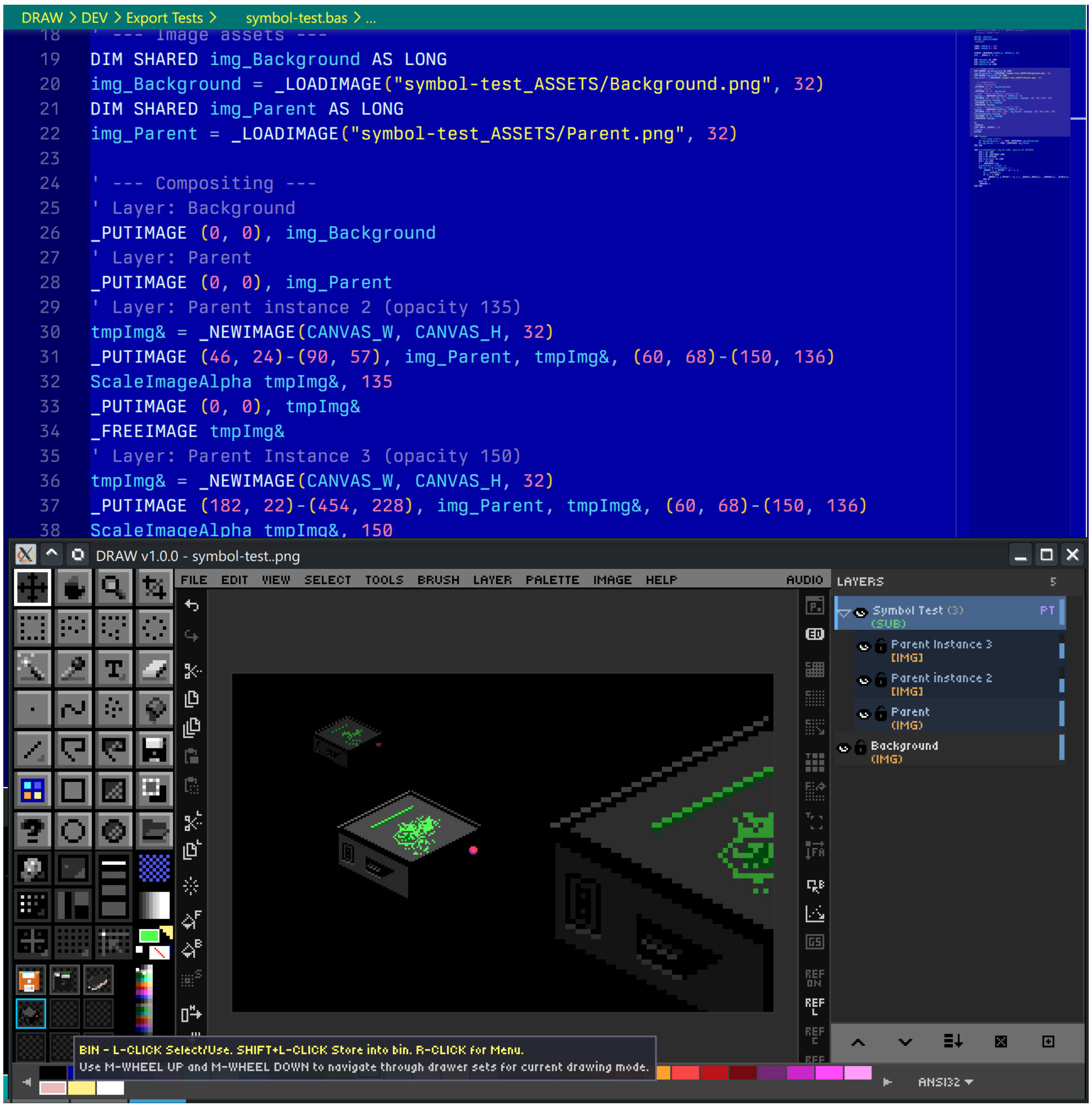
Symbol Layers

A **Symbol Parent** layer can have one or more **Symbol Children**. Drawing on the parent automatically syncs to every child — but each child can be **independently scaled and repositioned**. Use this for repeated elements like coins, enemies, tiles, or recurring UI components.

When you finish a child or want to have a child no longer inherit changes from it's parent:

- **Rasterize** — bake the child into a normal pixel layer (severs the link permanently).
- **Detach** — break the sync but keep the child as an independent layer.

Create them via `Layer → New Symbol Layer` or `Layer → Convert to Symbol`.



➡️ Next: Chapter 5 — Selection & Clipboard

Ch. 05 ✂️ Selection & Clipboard

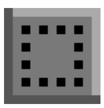
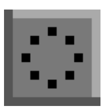
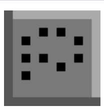
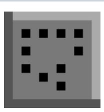
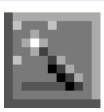
What you'll learn: Every selection tool DRAW offers, how selections clip drawing tools, the full clipboard vocabulary, and the Stroke Selection effect.

Selection Tools — Marquee, Freehand & Wand

🎯 **Goal:** Select regions for editing and manipulation.

A selection in DRAW does two things at once: it defines a region for clipboard operations, and it **clips every drawing tool** so paint only lands inside the marching-ants outline.

Marquee variants

Variant	Icon	What it does
Rectangle		Drag to draw a rectangular selection.
Ellipse		Oval / circular selection area.
Freehand		Draw the selection boundary by hand.
Polygon		Click each vertex; close with Enter.
Magic Wand		Click to select contiguous same-color pixels.

The shared modifier vocabulary works across **every** selection tool:

- `Shift` **+drag/click** — *add* to the existing selection (union).
- `Alt` **+drag/click** — *subtract* from the existing selection.

After a selection exists you get **resize handles** on its edges; **drag inside the marquee** to translate the selection (with `Move` semantics); **arrow keys** nudge by 1 pixel (10 with `Shift`).

Magic Wand specials

- `Shift+Click` — add to selection.
- `Alt+Click` — subtract.
- **Hold `F` and click** — instant flood fill with FG, no selection step. This works across ALL layers.
- **Hold `E` and click** — instant erase to transparent. This works across ALL layers.
- **Hold `W` and click** — sample from the merged composite of all visible layers, not just the active one.

Selection commands

Command	Shortcut
Select All	`Ctrl+A`
Deselect	`Ctrl+D`
Invert Selection	`Ctrl+Shift+I`
Expand / Contract	Image / Select menu
Selection from current layer	(non-transparent pixels)
Selection from selected layers	union mask

Copy, Cut, Paste & Clipboard Power

> 🎯 **Goal:** Master clipboard operations.

DRAW has more clipboard verbs than most editors because it cares about both layer-aware and pixel-aware semantics:

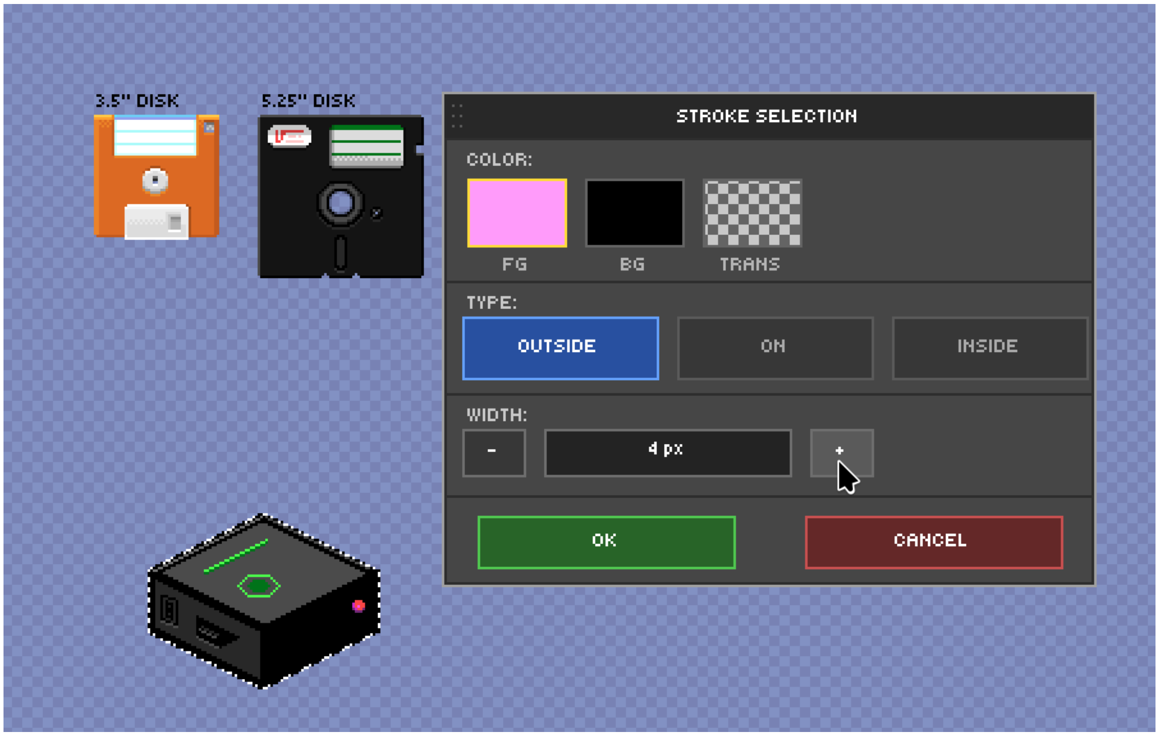
Action	Shortcut	Notes
Copy	`Ctrl+C`	Selection → clipboard.
Cut	`Ctrl+X`	Copy + clear.
Paste	`Ctrl+V`	Floats the clipboard **centered on the canvas** and **auto-engages the Move tool** — drag to reposition (Alt+drag stamps copies), then deselect / switch tool / `Enter` to commit.
Paste in Place	*Edit menu*	Paste at the *exact* original position.
Copy Merged	`Ctrl+Shift+C`	Copies the visible composite, not just the active layer.
Copy to New Layer	`Ctrl+Alt+C`	One-step "duplicate selection onto a new layer above".
Cut to New Layer	`Ctrl+Alt+X`	Same, but removes from the source.
Paste from OS Clipboard	`Ctrl+Shift+V`	Brings in any image you copied from another application.
Clear / Erase	`Ctrl+E` or `Delete`	Clears the selection (or the whole layer if no selection — no prompt).

Stroke Selection

`Edit → Stroke Selection` adds a stroke around the selection boundary. You can configure:

- **Width** in pixels.
- **Position** — inside, outside, or centered on the boundary.
- **Color** — current FG, or a custom value.

This is the cleanest way to outline finished sprites in a uniform thickness.



➡️ Next: Chapter 6 — Transforms & Image Adjustments

Ch. 06 🔄 Transforms & Image Adjustments

What you'll learn: Quick flips, rotations and scales, the on-canvas Transform overlay (Scale / Rotate / Shear / Distort / Perspective), the Move tool with Smart Guides, and DRAW's image-adjustment dialogs (Brightness/Contrast, Hue/Sat, Levels, etc.).

Quick Transforms — Flip, Rotate & Scale

🎯 **Goal:** Transform layers and selections quickly.

These are the no-dialog shortcuts you'll reach for hundreds of times a session.

Action	Shortcut	Icon
Flip Horizontal	`H`	
Flip Vertical	`Ctrl+Shift+H`	
Rotate 90° CW	`>`	
Rotate 90° CCW	`<`	
Scale Up 50%	`Ctrl+Shift+=`	
Scale Down 50%	`Ctrl+Shift+-`	

All of the above operate on the active selection if there is one, otherwise on the active layer (or the multi-selected layers). For canvas-level flips that move *every* layer at once, use the corresponding entries under the `Image` menu.

Transform Overlay — Scale, Rotate, Shear, Distort, Perspective

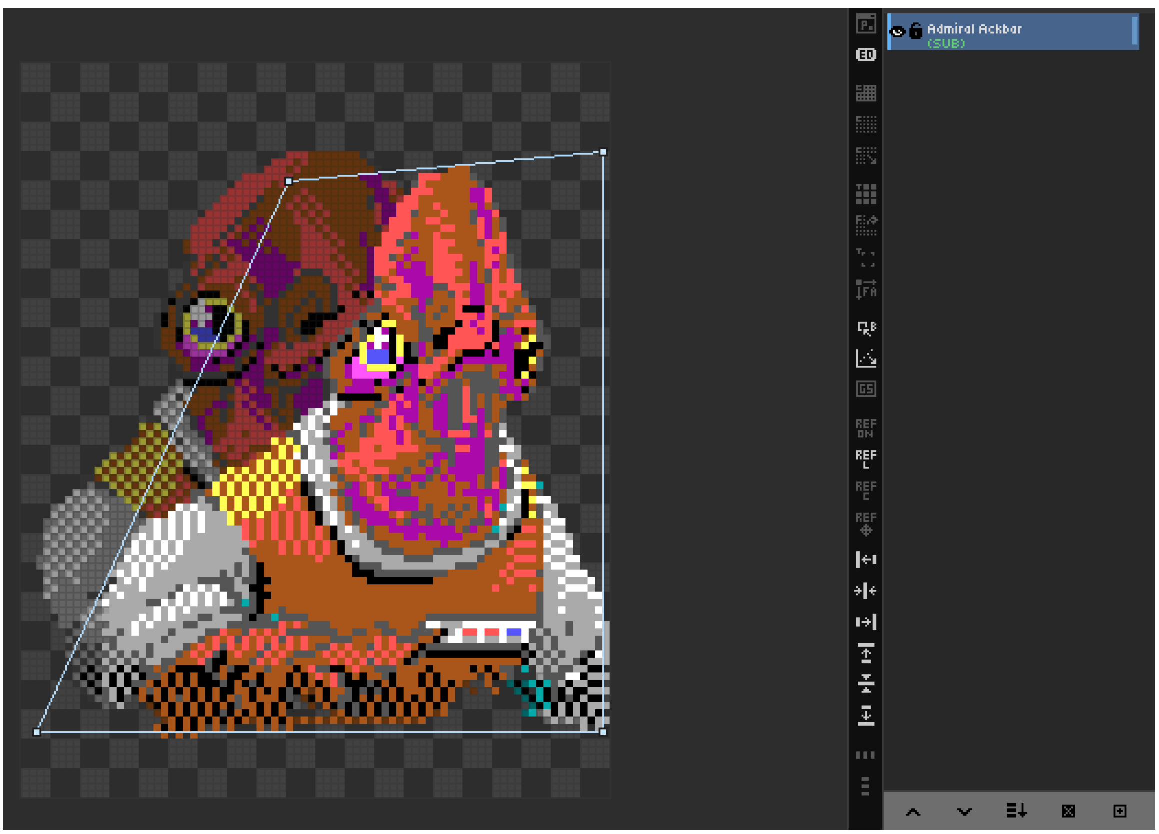
> 🎯 **Goal:** Use the interactive on-canvas transform tool.

Open it with `Edit → TRANSFORM…` (or via the Command Palette). Note that **Transform is not a toolbar tool** — it's a temporary overlay you commit or cancel.

The overlay supports five modes; switch between them with yhe menu bar items and use the corresponding hotkeys:

1. **Scale** — drag handles. `Shift` locks aspect ratio.
2. **Rotate** — drag *outside* the bounding box. `Shift` snaps to 15° increments.
3. **Shear** — drag the box edges to skew.
4. **Distort** — drag each corner independently.
5. **Perspective** — `Shift` enables a special wall / floor projection.

The overlay frame and handles are themeable in `THEME.CFG`. Press `Enter` to apply (full undo step is recorded), `Esc` to cancel without modifying anything.



Move Tool & Smart Guides

🎯 **Goal:** Reposition layer content precisely.

The **Move** tool (`V` ,) translates the active layer (or multi-selection). Modifiers:

- **Drag** — move.
- **Arrow keys** — nudge 1px (10px with `Shift`).
- **Alt+Drag** — clone stamp; the original stays where it was and the dragged duplicate becomes a new layer.
- **Ctrl+Arrows** — move selection content 1px (`Ctrl+Shift+Arrows` for a larger step; hold `Alt` to clone). *(Ctrl+Arrows resizes only while placing an imported/pasted image.)*
- **Shift+Click** — auto-select the topmost layer under the cursor.

Pro-tip: If you have an active selection, you can clone in Selection mode while holding `Ctrl+Alt`

Smart Guides

When you move a layer, DRAW can render **Smart Guides** — temporary alignment lines that snap to canvas edges, canvas centers, and other layers' edges. This makes pixel-perfect alignment between sprites trivial.

Toggle	Shortcut
Show Smart Guides	*Grid menu*
Snap to Smart Guides	*Grid menu*

Smart-guide colors and opacity are themeable.

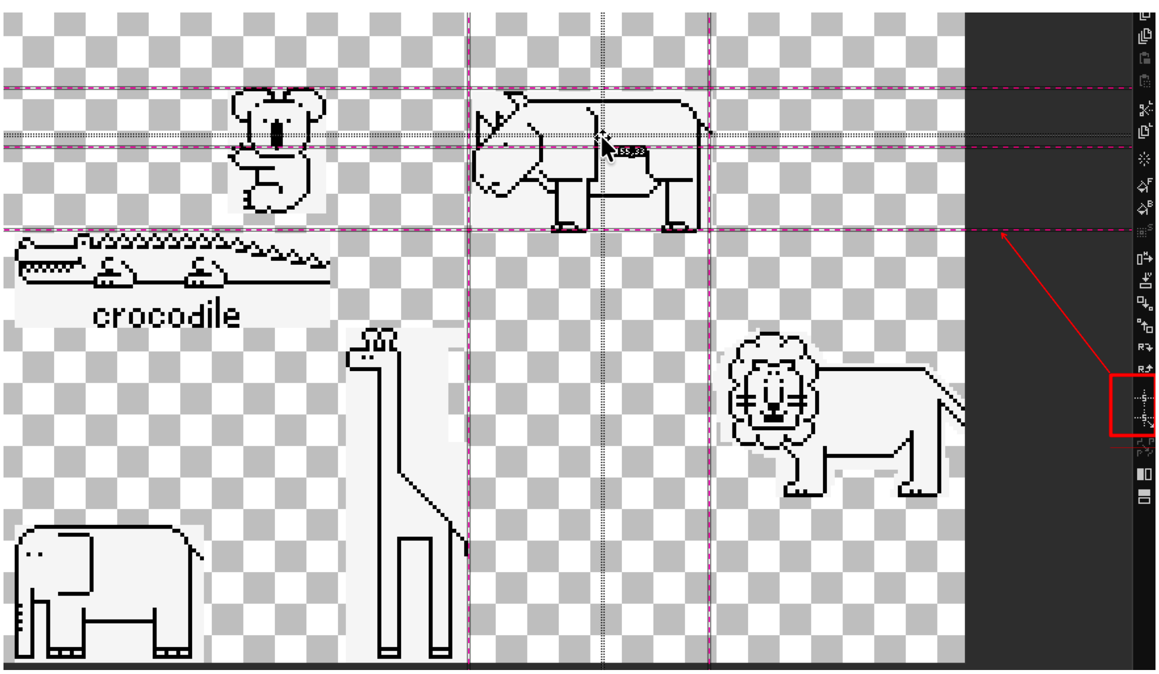


Image Adjustments — Color Correction & Effects

> 🎯 **Goal:** Apply per-layer color adjustments.

DRAW has two flavors of adjustment: **dialog-based** (with live preview, scrollable parameters) and **one-shot** (instant, undoable).

Dialog-based (live preview)

- **Brightness / Contrast** — Change the values and intensities of colors.
- **Hue / Saturation** — Rotate the colors on the color wheel and make them more or less colorful.
- **Levels** — Black point, White point, Gamma.
- **Color Balance** — Shadows / Midtones / Highlights.
- **Blur** — Gaussian, adjustable radius.
- **Sharpen** — adjustable intensity.

Hover any slider and use the **mouse wheel** for fine control.

One-shot

- **Invert** (RGB negative)
- **Desaturate** (luminosity grayscale) (note you can use `Ctrl+Shift+Alt+G` to toggle Gray Scale Mode preview ON or OFF)
- **Posterize** (N levels, with optional dithering)
- **Pixelate** (block size)
- **Remove Background**

Every adjustment **preserves the alpha channel** and is undone in a single `Ctrl+Z`.

The EFFECTS Menu — Photoshop / "Eye Candy" Style Effects

> 🎯 **Goal:** Stylize a layer with a large library of filters and generative effects.

Beyond the Image-menu adjustments, DRAW has a dedicated **EFFECTS** menu organized into **flyout submenu categories** (hover a category to open its list). Every effect opens a dialog with a **split ORIG / ADJ live-preview loupe**, **mouse-wheel sliders**, and an **OK / RESET / CANCEL** button row — **RESET** restores that effect's default parameters. Effects apply per-layer (multi-layer aware), preserve alpha where appropriate, and undo in a single `Ctrl+Z`.

- **ADJUST** → Gamma, Sepia, Threshold, Colorize, Gradient Map, Solarize, Duotone / Tritone
- **STYLIZE** → Glow, Film Grain, Vignette, Outline / Stroke, Edge Detect (Sobel), Grow / Shrink (dilate / erode), Chromatic Aberration, **Emboss** (with a **Light Angle**), Wind
- **LAYER FX** → **Drop Shadow** (Distance / **Angle** / Softness / Opacity), Perspective Shadow (Length / **Cast Angle**), **Bevel** (Height / Strength / **Light Angle** / Direction), Outer Glow, Inner Glow, Chrome / Metallic
- **DISTORT** → Wave / Ripple, Twirl, Pinch / Bulge (spherize), Kaleidoscope (auto mandala)

- **PIXELATE** → Crystallize, Stained Glass, Mosaic / Tessellate, Extrude, Pointillize
- **NOISE** → Add Noise, Median / Despeckle, Dust & Scratches
- **RENDER** → Clouds, Difference Clouds, Lens Flare, Terrain, Grid (with perspective), Sky (day / night / space)

SHAPE → (Eye Candy "Shape", 20 effects)

Effects that work off the shape's alpha edge — backlights, glows, shadows, and elemental overlays: **Backlight, Corona, Cutout, Bevel, Chrome, Extrude, Outer Glow, Inner Glow, Drop Shadow, Perspective Shadow, Motion Trail, Rust, Snow, Fire, Smoke, Icicles, Drip, Glass, Electrify, Water Drops.**

TEXTURE → (Eye Candy "Texture", 15 effects)

Procedural surfaces painted onto the shape, each with a **MIX** slider (0–100) to blend the texture with the original art instead of replacing it outright: **Wood, Marble, Brick Wall, Brushed Metal, Weave, Clouds, Swirl, Water Drops, Ripples, Animal Fur, Texture Noise, Stone Wall, Reptile Skin, Diamond Plate, Lightning.**

> 💡 Effects that draw *outside* the shape (Drop Shadow, Outer Glow, Fire, Smoke, Icicles, Drip, Backlight, Corona, Motion Trail, Snow flecks) work best on a **transparent layer** so there's room around the artwork — add one with `Ctrl+Shift+N`. Effects that use a colour (Backlight, Corona, Inner/Outer Glow, Gradient Map, Duotone, Drop/Perspective Shadow) read the current **FG / BG** colours.

➡️ Next: Chapter 7 — Text System

Ch. 07 📝 Text System

> **What you'll learn:** How DRAW's text tool works, the bundled and custom font support, rich per-character formatting, persistent text layers, and the unique **Character Mode** for ANSI-style art.

Text Tool Basics — Fonts & Entry

> 🎯 **Goal:** Add text to your pixel art.

Press `T` to grab the Text tool, then click on the canvas to drop a text caret. Type to enter characters; `Enter` starts a new line; `Escape` commits the text and exits text-entry mode.

Bundled fonts

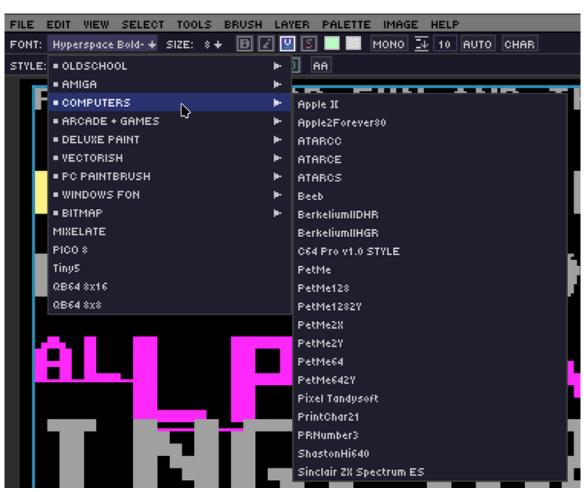
Font	Shortcut	Description
VGA	`T`	Classic 8×8 monospace — the iconic DOS look.
Tiny5	`Shift+T`	Minimal 5×5 — perfect for tight UI labels.
Custom TTF / OTF	`Ctrl+T`, or Left-Click in the font dropdown	Load any vector font from disk.

Bitmap fonts

DRAW ships with **24 color bitmap fonts** in the DPaint-style `.bmp` spritesheet format. Their pixels are preserved verbatim — no antialiasing, no recoloring — at a fixed native height per font. PSF, BDF, and Fontaption formats are also supported.

The Text Property Bar

When the text tool is active, a property bar appears with the font dropdown, size, and the standard formatting toggles (bold, italic, underline, strikethrough). This bar is fixed in position under the menu bar and not movable or undockable.



Rich Text — Per-Character Formatting

> 🎯 **Goal:** Apply advanced text formatting.

While editing a text layer, every character can carry its own properties. Selection works exactly like a desktop word processor:

Action	Result
`Shift`+Arrow	Select characters one at a time.
Double-click	Select word.
Triple-click	Select line.
Quad-click	Select all.
`Ctrl+A`	Select all.

Per-character properties:

- **Bold / Italic / Underline / Strikethrough**
- **Outline** — color + size.
- **Shadow** — color + X/Y offset (1–10px).
- **Per-character FG and BG colors**.
- **Letter spacing** and **line height**.

Auto-wrap mode and style presets

Auto-wrap reflows the text inside the bounding box. Style presets let you save the entire formatting state of a selection as a named preset and apply it elsewhere with one click — Save / Load / Update / Delete buttons live in the property bar.

Text-local undo

While in text-entry mode, **`Ctrl+Z` and `Ctrl+Y` operate on a separate 128-state text-local history**. This means typo experiments don't pollute your global history. Sound feedback on each keystroke is configurable in audio settings.

Style Auto-Sync

As you move the text cursor, the formatting toolbar now updates live to reflect the style of the character at the cursor. This makes it effortless to inspect and modify mixed-style text — the toolbar always shows the current character's formatting.

Text Layers & Character Mode

> 🎯 **Goal:** Master persistent text and ANSI-style art.

Text layers

Once you commit a text-tool entry, it becomes a **text layer** — fully persisted in `.draw` files and re-editable forever. Click it again with the text tool to resume editing. From the Layer menu you can:

- Rasterize a single text layer (bake it into pixels).
- Rasterize all text layers (one-shot flatten of every text layer).
- Insert a new empty text layer ready for typing.

Character Mode

Character Mode is for **ANSI / DOS / text-mode art**. Once enabled, the canvas is treated as a grid of cells with a virtual cursor:

> NOTE: DRAW is NOT a full fledged text mode art program, but this is a neat way to make GUI elements!

- Free grid navigation with arrow keys.
- `F1` – `F12` insert ANSI block characters (`░ ▒ ▓ █ ▀ ▄ ▌ ▐`).
- "Font stickiness" — the chosen font follows the cursor as you navigate.
- The DOT and RECT drawing tools fill cells with the current glyph instead of pixels.
- `Alt+U` picks both the FG and BG colors from the character under the cursor.
- A character grid overlay snaps cursor positions cleanly.

Character Map panel

Open with `Ctrl+M` for a 16×16 grid of all 256 glyphs in the active font.

- Click a glyph to insert it (or use it as a custom brush).
- Dock the panel left or right.
- Toggle Unicode / CP437 mapping from the **View menu** → **Charmap Unicode/CP437 Toggle** (or the command palette).

> 📸 **Screenshot needed — Character Map panel**
>
> - **Setup:** Switch to VGA font, enable Character Mode, open `Ctrl+M`.
> - **Action:** Hover one of the block-shading glyphs (e.g., `▒`).
> - **Capture:** Character Map panel docked right, glyph grid clearly visible.
> - **Save as:** `images/ch07-charmap.png`

> 🎨 **Try it — DOS-style title card**
>
> 1. Enable Character Mode.
> 2. Pick the VGA font.
> 3. Use `F1`–`F4` to lay block shading along the borders.
> 4. Switch FG/BG colors per cell with `Alt+U` to match an existing color set.

➡️ Next: Chapter 8 — Grid, Symmetry & Drawing Aids

Ch. 08 📐 Grid, Symmetry & Drawing Aids

> **What you'll learn:** How to use DRAW's four grid geometries, three symmetry modes, angle-snap, the crosshair assistant, Pixel Perfect mode, and the pattern tile preview.

Grid System — 4 Geometry Modes

> 🎯 **Goal:** Use grids for precise pixel art layout.

Action	Key
Toggle grid visibility	`'` (apostrophe)
Toggle pixel grid (≥ 400% zoom)	`Shift+'`
Toggle snap-to-grid	`;` (semicolon)
Increase grid size (uniform W+H)	`.`
Decrease grid size (uniform W+H)	`,`
Toggle grid alignment (Corner ↔ Center)	`/` (slash)
Cycle geometry mode	`Ctrl+'`
Set grid size from brush / custom brush	`Ctrl+Shift+/`

Grid sizes range from 2 to 256 pixels per cell.

Independent axis sizing

Hold `G` and press an arrow key to resize width or height separately — useful for non-square grids (e.g., a 16×8 isometric tile cell):

Key	Effect
`G + Right`	Increase grid **width** by 1px
`G + Left`	Decrease grid **width** by 1px
`G + Down`	Increase grid **height** by 1px
`G + Up`	Decrease grid **height** by 1px

The `.` / `,` keys adjust width and height together; `G+Arrow` adjusts each axis independently.

Geometry modes

1. **Square** (default) — standard tile grid.

2. **Diagonal** — 45° rotated diamonds.
3. **Isometric** — 2:1 pixel-art standard.
4. **Hexagonal** — flat-top hexagons.

Alignment

Each geometry can be set to either **Corner alignment** (snap on intersections) or **Center alignment** (snap on cell centers). The choice depends on whether you want to draw *between* cells or *on* them.

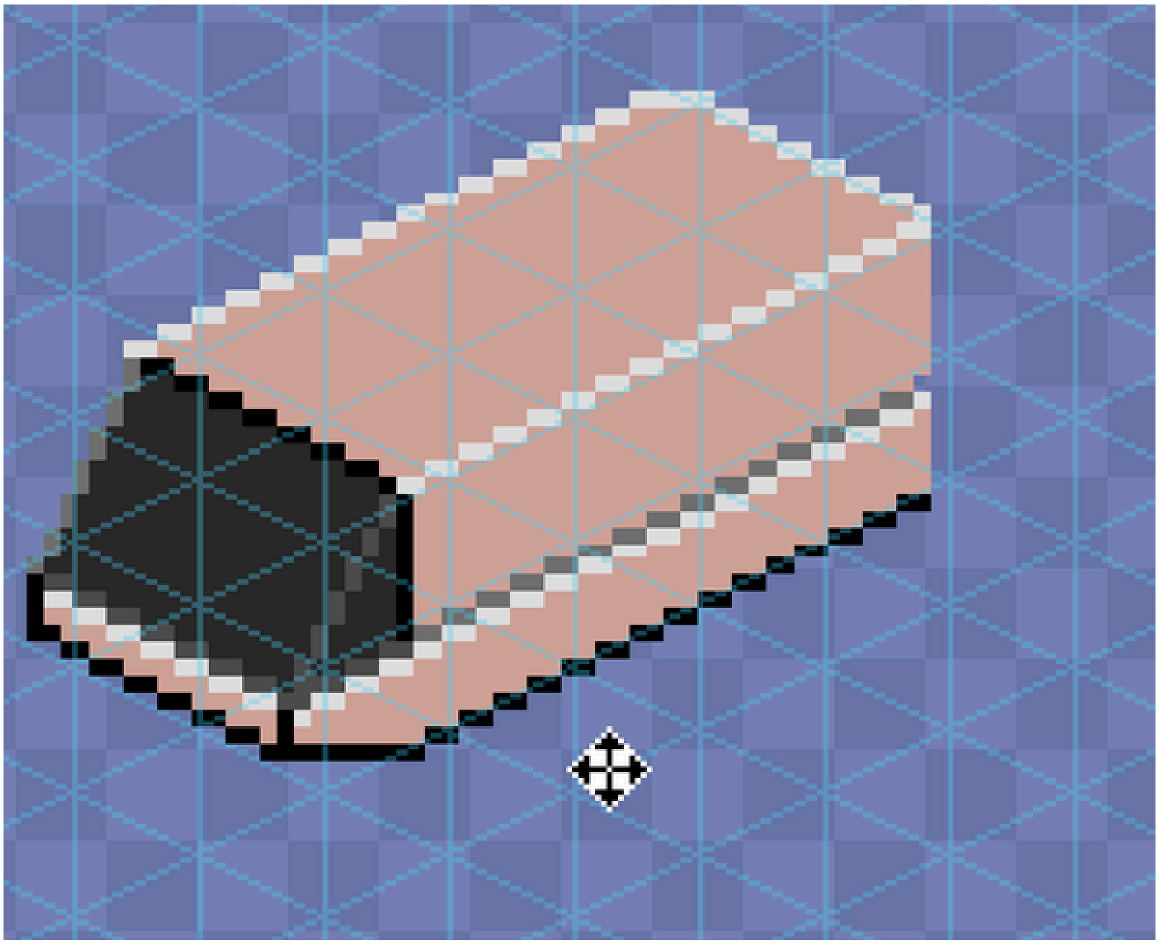
Grid Cell Fill

When grid cell fill is enabled, the fill tool paints whole cells of the active geometry — squares for the square grid, diamonds and triangles for diagonal, and hexagons for hex.

`Ctrl+Shift` while drawing temporarily **bypasses snap**, so you can sketch freely without disabling the grid. All grid state is persisted in `.draw` files.

🎨 **Try it — isometric cube**

1. `Ctrl+'` to switch to Isometric mode.
2. `;` to enable snap.
3. Use the Polygon tool to outline the top, left, and right faces of a cube on three layers.
4. Apply Multiply/Screen blend modes for top vs. side shading.



Symmetry Drawing — Mirror & Kaleidoscope

> 🎯 **Goal:** Create symmetrical art effortlessly.

Press `F7` to cycle symmetry modes:

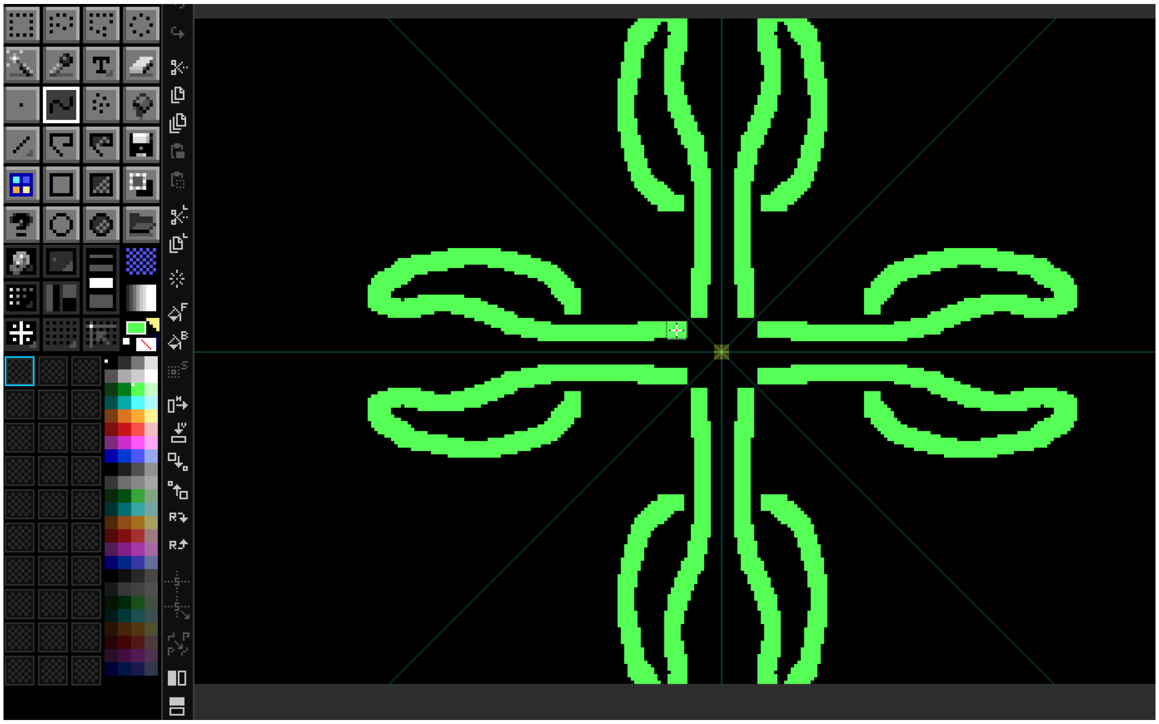
Mode	Status bar	Copies	Shape
Off	`SYM:0`	1	—
Vertical	`SYM:1` (|)	2	Bilateral
Cross	`SYM:2` (+)	4	Quad
Asterisk	`SYM:3` (*)	8	Kaleidoscope

`Ctrl+Click` repositions the symmetry center anywhere on the canvas. DRAW renders subtle visual guides at the center.

Symmetry works with **every** drawing tool — Brush, Dot, Line, Rect, Ellipse, Polygon, Spray, Custom Brush. `F8` disables symmetry (or, in Fill Adjust contexts, opens the adjust overlay).

🎨 **Try it — 8-way mandala**

1. New 256×256 canvas.
2. `F7` × 3 until status bar reads `SYM:3`.
3. `Ctrl+Click` the canvas center.
4. Draw one curved stroke from center outward — eight reflections appear instantly.
5. Add layers and blend modes to taste.



Drawing Aids — Angle Snap, Crosshair & Assists

> 🎯 **Goal:** Use helpers for clean, precise artwork.

Angle Snap

Hold `Ctrl+Shift` while drawing a Line, Polygon, Brush stroke, or Dot connecting line. DRAW supports two angle-snap regimes:

- **Degree mode** — 15° / 30° / 45° / 90°.
- **Pixel-art mode** — integer-ratio angles like 1:1, 2:1, 1:2, 3:1, etc., which produce visually clean stairsteps without the half-pixel artefacts of 30° / 60° lines.

Choose the mode in `DRAW.cfg`.

Crosshair Assistant

Hold `Shift` to display a full-screen crosshair through the cursor — invaluable for aligning distant elements. Color, opacity, width, and the optional outline stroke are all themeable.

Pixel Perfect mode

`F6` toggles **Pixel Perfect** brush smoothing — DRAW retroactively removes L-shaped corners from your strokes for the cleanest possible 1-pixel outlines.

Other view aids

- **Grayscale Preview** — `Ctrl+Alt+Shift+G`. View the composite as luminance only to check value structure.
- **Pattern Tile Mode** — *View → Pattern Tile Mode* (or the tile-mode button in the Advanced Bar). Renders the canvas as a 3×3 tiled preview so seamless textures show their seams immediately.
- **Canvas Border** — `#`. Toggles a thin border around the canvas (useful at high zoom over dark themes).

➡️ Next: Chapter 9 — Custom Brushes & Drawer Panel

Ch. 09 🪄 Custom Brushes & Drawer Panel

What you'll learn: How to capture, transform, recolor and stamp custom brushes; the 30-slot Drawer panel for brush / pattern / gradient libraries; and DRAW's six dithering algorithms.

Custom Brushes — Capture, Transform & Paint

🎯 **Goal:** Create and use custom brushes.

A **custom brush** is any selection captured into the brush slot — DRAW remembers the pixels and treats them as the active stamp for every drawing tool that supports stamping.

Capturing

Make a selection, then capture it as a brush via the Brush menu, the Organizer button, or the Command Palette. There are two capture modes:

- **From Current Layer** — only non-transparent pixels.
- **From Selected Layers** — union mask across the multi-selection.

Non-rectangular shapes are preserved via the alpha channel.

Transform controls

Action	Key
Flip Horizontal	`Home`
Flip Vertical	`End`
Scale Up	`PgUp`
Scale Down	`PgDn`
Reset Scale	`/`

Modes that work *with* a custom brush

Tool	Effect
Brush / Spray	Stamp the brush at every step.
Line / Rect / Ellipse / Polygon	Stamp along the geometry.
Fill	**Tiled fill** of the brush as a seamless pattern.
Eraser	Stamp transparency in the brush shape.

Recolor and outline modes

- **Recolor** (`F9`) — paints the brush in the current FG color, preserving its alpha.
- **Outline** (`Shift+O`) — adds a BG-colored outline around the brush silhouette.

Export

`F12` exports the current brush as a standalone PNG (alpha preserved) — perfect for sharing brushes between projects.

> 🎨 **Try it — leaf scatter**
>
> 1. Draw a small 8×8 leaf on a transparent layer.
> 2. Marquee around it; capture as custom brush.
> 3. Switch to Spray, increase brush size, set FG to a green family.
> 4. Spray a tree's foliage onto a new layer in seconds.

Drawer Panel — 30 Reusable Slots

> 🎯 **Goal:** Organize and reuse brushes and patterns.

The **Drawer panel** is a docked grid of 30 slots that hold reusable brushes, patterns, or gradients. Each slot is one click away from being the active stamp.

Drawer modes

Mode	Shortcut	Use for
Brush	`F1`	Stamping and painting.
Gradient	`F2`	Color transitions for fills.
Pattern	`F3`	Seamless tiled fills.

Storing and managing slots

Action	Result
`Shift` +Left-click an empty slot	Store the current brush there.
Right-click any slot	Context menu — load, save, clear, replace, export, etc.
Drag-and-drop slots	Reorder.
Drop image files onto the panel	Batch import as slot brushes.

`.dset` import / export

The drawer state can be saved as a `.dset` file and reloaded later — a clean way to share brush packs and pattern libraries between projects or with other artists. There's a sample set in `DEV/brush-set.dset`.

Mini palette

The Drawer also hosts a **mini palette** for quick FG / BG selection without hopping back to the main palette strip.

1-bit patterns

Patterns can be stored with **opaque backgrounds** (1-bit black/white) so that flooding with a 1-bit pattern under a colored FG produces classic Aldus PageMaker / DPaint dither patterns.

Paint modes

The drawer ties into three paint modes that affect Fill and Brush behaviour:

- **Normal** — solid current FG color.
- **Pattern** — tile the active drawer slot.
- **Gradient** — interpolate between palette stops via the active gradient slot.

Dithering algorithms

When painting gradients (or applying Posterize) you can pick from:

- **Ordered** — Bayer 2×2, 4×4, 8×8.
- **Floyd-Steinberg** — error diffusion.
- **Atkinson** — classic Mac dither.
- **Stucki** — heavier diffusion.
- **Blue Noise** — perceptually uniform.

➡️ Next: Chapter 10 — File I/O & Export

Ch. 10 💾 File I/O & Export

> **What you'll learn:** Every way to get artwork in and out of DRAW — the native `.draw` format, nine raster export formats, the famous QB64 source-code export, and the Extract Images sprite-decomposition tools.

Opening & Saving — Formats Explained

> 🎯 **Goal:** Understand all file format options.

Open / Import

Format	Shortcut / menu
Image (PNG / BMP / JPG / GIF)	`Ctrl+O`
Project (`.draw`)	`Alt+O`
Aseprite (`.ase` / `.aseprite`)	File → Open Aseprite
Photoshop (`.psd`)	File → Open Photoshop
Import Image (oversized, interactive placement)	File → Import Image
Drag-and-drop file onto window	Windows
Command-line argument	`DRAW path/to/file.draw`

Save options

Command	Shortcut	Notes
Save	`Ctrl+S`	Silent resave to current path.
Save As...	`Ctrl+Shift+S`	Prompt for new name and format.
Export Selection	`Ctrl+Alt+Shift+S`	Save just the selected region.
Export Layer as PNG	Layer menu	One-click per-layer PNG.
Export Brush as PNG	`F12`	Current custom brush.

The **Recent Files** list holds up to 10 entries with quick-jump shortcuts `Alt+1` ... `Alt+0`.

`File → New from Template…` reads from `ASSETS/TEMPLATES/` and gives you a starter canvas with appropriate dimensions, palette, and grid for common targets (NES sprite, Game Boy tile, Lospec sheet, etc.).

The `.draw` Format — PNG with Superpowers

> 🎯 **Goal:** Understand the native DRAW project format.

A `.draw` file is a **valid PNG file** with a custom `drAw` chunk embedded inside. Every image viewer on the planet can render the flattened preview. Only DRAW can read the rich data inside the chunk and reconstruct the editable project.

What `.draw` preserves:

- All layers, blend modes, opacities, names, groups, symbol parents/children.
- Active palette state and the auto-generated `[DOCUMENT]` palette.
- Tool states, grid settings, snap, crosshair config.
- Reference image configuration and overlay opacity.
- Text layer data (re-editable!).
- Extract Images settings.
- Character Mode state.

The format is versioned for forward compatibility — newer DRAW builds still read older `.draw` files, and they extend the chunk safely when new features arrive.

Export As — 9 Image Formats

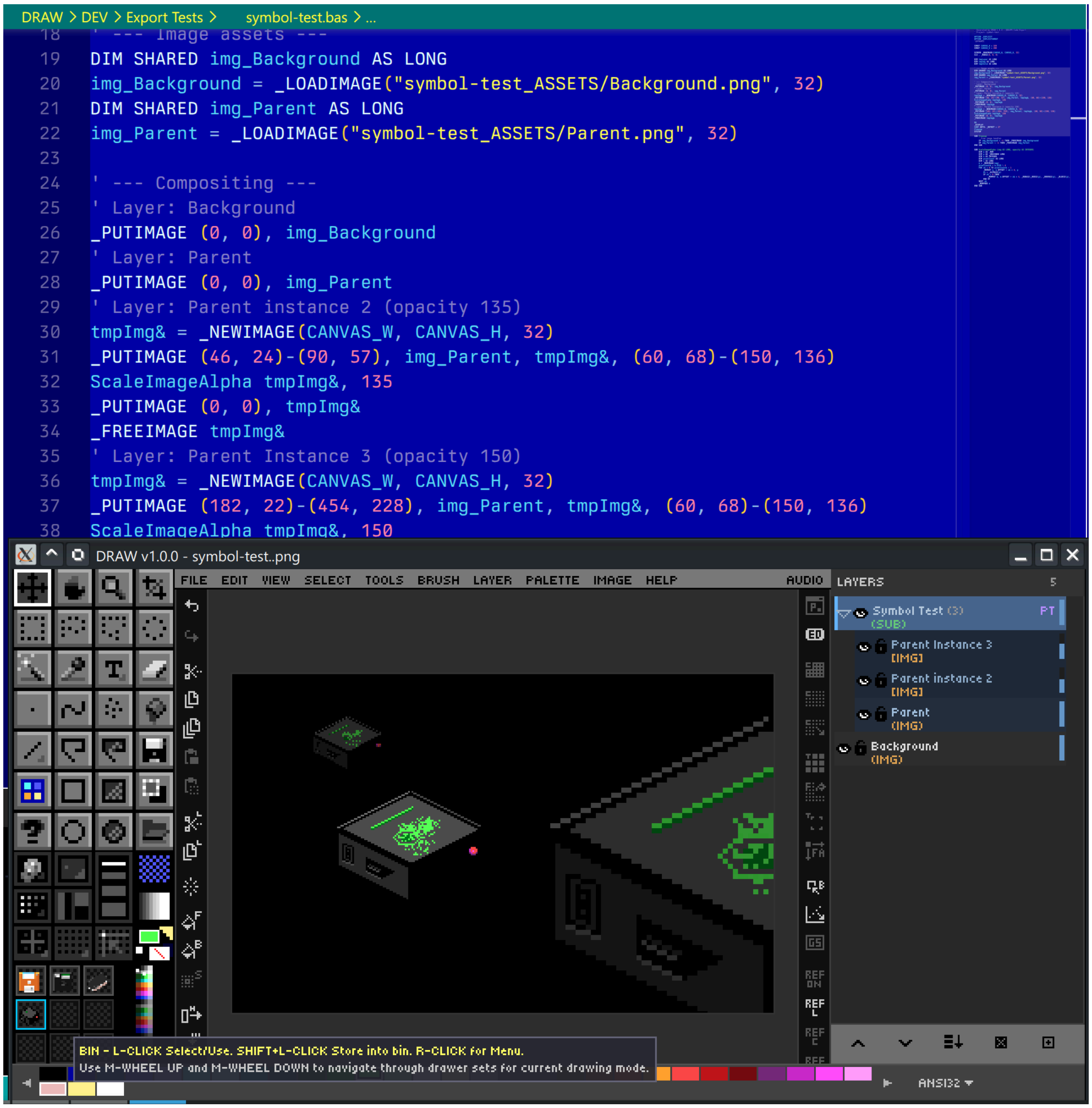
> 🎯 **Goal:** Export artwork in various formats.

Format	Use case
PNG (native `.draw`)	Embeds the full project in a PNG; viewable anywhere, re-editable in DRAW.
PNG (plain)	Standard PNG, no metadata.
GIF	Static, palette-aware.
JPEG	Lossy compression for previews.
TGA	Truevision; popular in game dev.
BMP	Windows Bitmap — for retro tooling.
HDR	High Dynamic Range.
ICO	Windows Icon (multi-resolution).
QOI	Modern lossless Quite OK Image format.

QB64 Source Code Export

The signature feature: export your artwork as a self-contained `.bas` program. The file declares a screen, sets pixels in order, and runs as a standalone QB64-PE compile. Open it, hit compile, and your sprite paints itself onto the screen. This is wonderful for:

- BASIC tutorials.
- Embedding small graphics in QB64-PE programs without shipping image files.
- Showing how a piece of pixel art was constructed pixel by pixel.



ANSI Art — Import & Export

> 🎯 **Goal:** Bring classic ANSI/BBS art into DRAW as pixels, and turn your pixel art into shareable ANSI.

DRAW reads and writes the ANSI-art family of formats, so you can drop lo-fi mockups into DRAW, refine them as pixels, and hand them back out as `.ans`.

Importing ANSI (File → Import ANSI...)

Supported inputs:

Kind	Extensions	Notes
Text ANSI	`.ans` `.asc` `.nfo` `.diz`	SGR escape sequences — 16-color, **iCE**, **256-color** (`38;5;n`), and **24-bit truecolor** (`38;2;r;g;b`). UTF-8 block/box art is auto-detected and mapped to CP437.
BIN	`.bin`	Raw character+attribute pairs; width from SAUCE (`BinaryText`).
XBIN	`.xb` `.xbin`	Full XBIN: custom 16-color palette, embedded font, and RLE compression.
TundraDraw	`.tnd`	24-bit truecolor binary stream.

The **Import ANSI** dialog previews the art and its detected format, and lets you choose how it rasterises onto the canvas:

- **WIDTH** — `8 PX` (square) or `9 PX` (authentic VGA 9-dot; box-drawing characters join across cells).
- **ASPECT** — `SQUARE` or `DOS 1.2` (the classic non-square CRT stretch).
- **SCALE** — `1×` – `8×` nearest-neighbor.
- **Cell height** follows the SAUCE font: `IBM VGA` → 8×16, `VGA50` / `EGA43` → 8×8.

If an **XBIN** carries an embedded **font** or a custom **palette**, the dialog offers to install them into your libraries:

- **Import font** writes the font to `FONTS/ANSI Imports/` so it appears as a bitmap font for the Text tool.
- **Import palette** writes a `.GPL` into your user palettes and refreshes the picker.

You can also open ANSI art straight from the command line: `./DRAW.run artwork.xb`.

Exporting ANSI (File → Export ANSI...)

Export converts the canvas to **half-block** ANSI (each character cell shows two stacked pixels via `▀`). The dialog gives a live source-vs-render preview plus:

Control	Choices
FROM	`DOCUMENT` (flattened) · `LAYER` (current) · `SELECTION` (marquee region) — auto-selects Selection when one is active
COLOR	`16` · `iCE` · `256` · `RGB` (24-bit) — auto-detected from the art
CELLS	`PER-PIXEL` (one canvas pixel per cell — maximum detail) · `1:1 8PX` (one 8-px cell — round-trips terminal art at native size)

FONT	`VGA 8x16` · `VGA50 8x8` (written to the SAUCE record)
SIZE	Column count (auto, or stepped)

A byte-exact **SAUCE** record (dimensions + font) is appended. Use **Selection** as the source to mock a scene in pixels and export just one section to ANSI.

> 💡 **Round-trip tip:** `PER-PIXEL` keeps every pixel but renders ~8× larger in a terminal/viewer; `1:1 8PX` maps 8×16-pixel blocks to one cell, so exported terminal art re-imports at its original size.

Extract Images — Sprite Sheet Decomposition

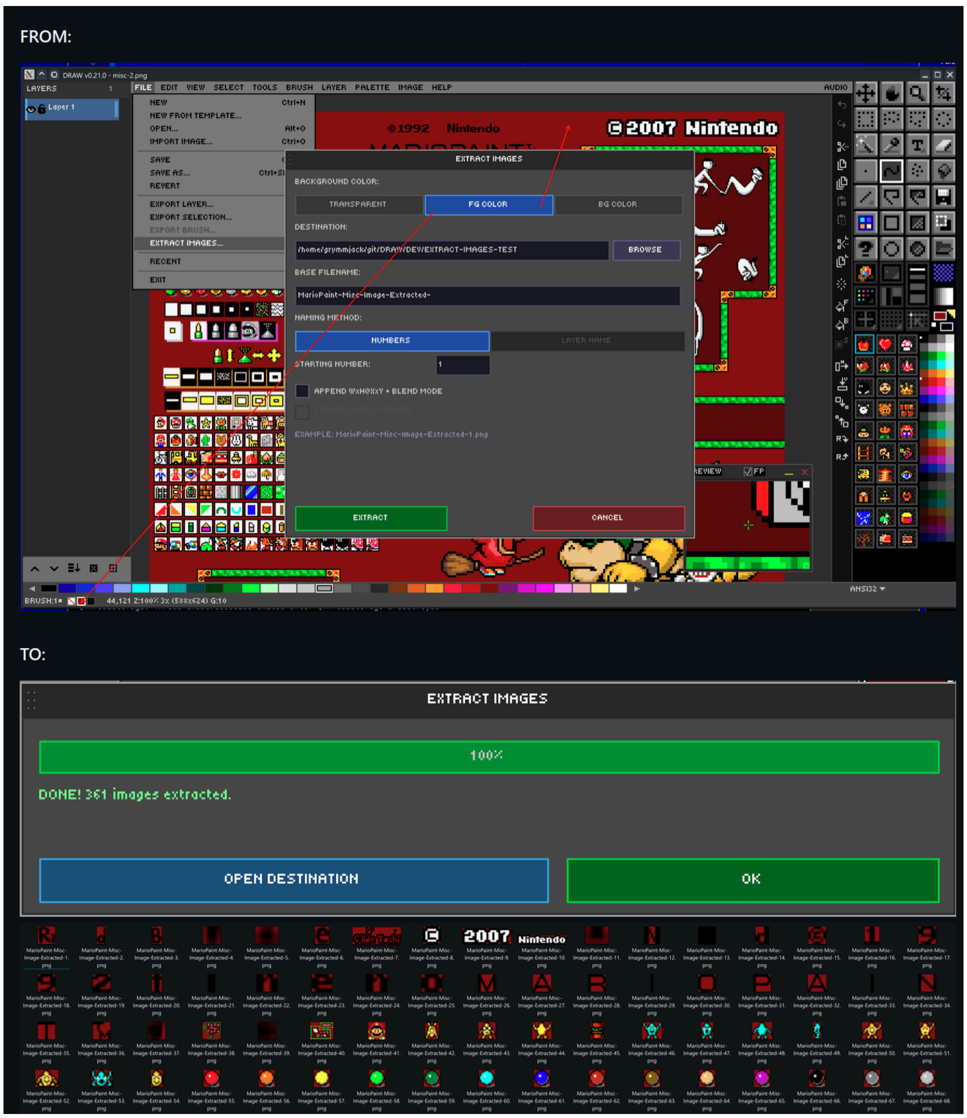
🎯 **Goal:** Extract sprites from sheets or compositions.

DRAW can decompose a finished image into individual sprite PNGs. Three extraction methods are available:

- **Flood-fill connected regions** — auto-detect distinct sprites by transparent gutters. Best for unsorted sprite sheets.
- **Per-layer extraction** — each layer becomes one PNG. Best when your sheet was built layer-by-layer.
- **Merged extraction** — flatten visible layers and extract regions from the result.

Background options for each output sprite: **Transparent**, **FG color**, or **BG color**. Output is a folder of separate PNG files; the settings you pick are persisted in the `.draw` file so you can re-export later with one click.

There is a related command, **Extract From Grid...**, which slices a regular grid (e.g., 16×16 cells) into named PNGs — perfect for tilesets — and **Extract To Layers From Grid...**, which slices a grid into named *layers* in a new project.



➡️ Next: Chapter 11 — Canvas & View Controls

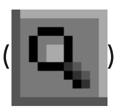
Ch. 11 🖥️ Canvas & View Controls

> **What you'll learn:** How to navigate any canvas — zoom, pan, the floating Preview Window, pattern-tile preview, reference images, fill / clear shortcuts, and canvas resize / crop.

Zoom, Pan & Navigation

> 🎯 **Goal:** Navigate canvases of any size efficiently.

Zoom

Action	Shortcut
Zoom in/out continuously	`Ctrl`+Mouse wheel
Zoom in / out 1 step	`Ctrl+=` / `Ctrl+-`
Reset to 100%	`Ctrl+0`
Zoom to canonical level	Hold `Z` and press `1-8` for 100%-800%, `9` for 1600%, and `0` for 3200%
Zoom tool	`Z` ()
Zoom to dragged region	`Z` then drag
Zoom out (Zoom tool)	`Alt`+Click

Zoom snaps to canonical levels: 25%, 50%, 75%, 100%, 200%, 300%, 400%, 600%, 800%, 1600%, 3200%

Pan

You can pan **with any tool active**, no need to switch:

- Middle-mouse-drag.
- `Spacebar`+drag.
- **Double-middle-click** — instantly resets pan and zoom to fit the canvas.

Scrollbars appear automatically for canvases larger than the viewport. The canvas border (`#`) outlines the edge of the canvas at any zoom.

Preview Window, Tile Mode & View Options

> 🎯 **Goal:** Use preview and tiling for workflow efficiency.

Preview Window

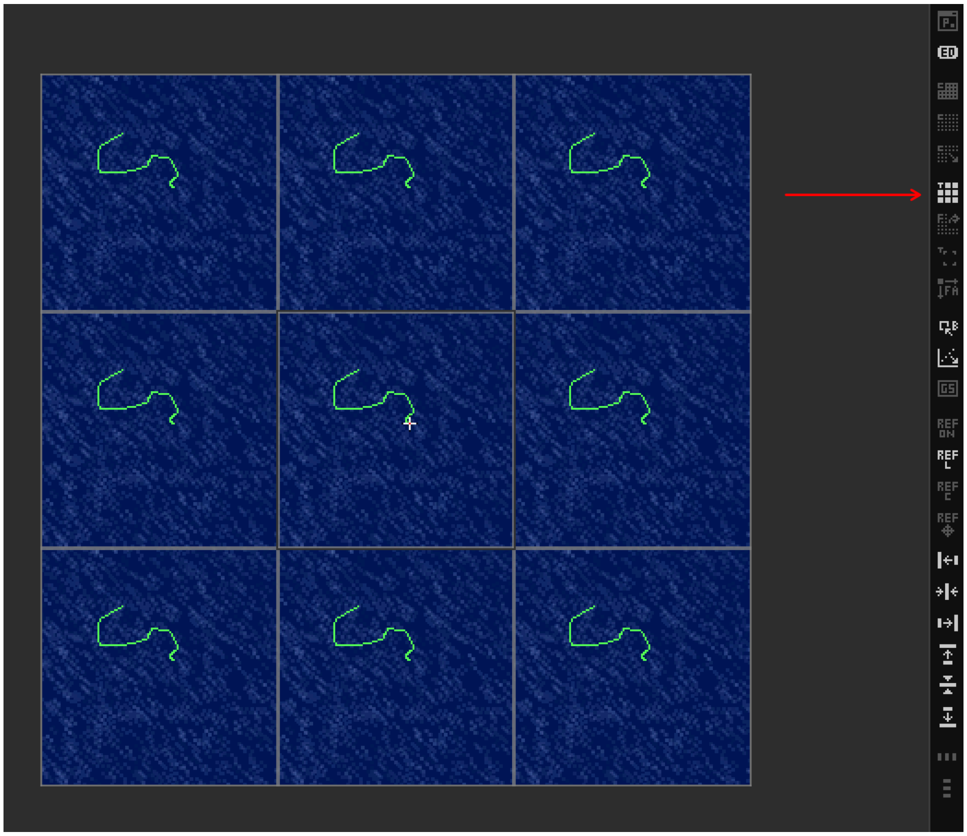
`F4` toggles the **Preview Window**, a floating, resizable, repositionable secondary view. It runs in one of three modes:

1. **Follow** — a magnifier that tracks the cursor.
2. **Floating Image** — a static reference display (e.g., the source you're tracing).
3. **Bin Quick Look** — hover any drawer slot to see its content full-size in the preview.

Independent zoom and pan inside the preview, plus `Alt` **+Click to color-pick** without switching tools. The preview also remembers up to 10 recent images for quick swaps.

Pattern Tile Mode

View → Pattern Tile Mode (or the tile-mode button in the Advanced Bar) toggles a 3×3 tiled preview (up to 512×512 source). Seams in seamless textures jump out instantly because they appear nine times.



Other view options

- **Grayscale Preview** — `Ctrl+Alt+Shift+G`. Inspect value structure without color distraction.
- **Reference Image** — `Ctrl+R`. See Chapter 15.

Canvas fill & clear

- `Delete` — clear (Edit → Clear). Removes the active selection's pixels (or the whole layer if no selection). No prompt.
- `Backspace` — fill with FG color. Instant, no prompt.
- `Shift+Backspace` — fill with BG color. Instant.

Canvas crop & resize

- **Crop tool** () — interactive with handles; commit with Enter.
  - Drag handles **inward** to trim the canvas to the selected region.
  - Drag handles **outward past a canvas edge** to **grow** the canvas in that direction — new pixels are transparent on all layers. This is a quick alternative to the canvas resize dialog when you just need to add space on one side.
  - Arrow keys nudge the crop box 1px; `Shift+Arrow` nudges 10px.
- **Image → Canvas Resize...** — dialog with anchor selection and pixel/percent toggle.

➡️ Next: Chapter 12 — UI Customization & Settings

Ch. 12 ⚙️ UI Customization & Settings

> **What you'll learn:** How to configure DRAW to your taste — the eight-tab settings dialog, theming, panel docking, and the various ways `DRAW.cfg` works under the hood.

Settings Dialog — All 8 Tabs Explained

> 🎯 **Goal:** Configure DRAW to your preferences.

Open the settings dialog with `Ctrl+,` (comma) or `Edit → Settings`. The dialog has eight tabs covering every persistent option.

Tab	What it controls
General	Display scale, fullscreen toggle, FPS limit, UI scaling, **Enable AI Features**.
Grid	Default grid size, geometry, alignment, snap state, crosshair appearance.
Palette	Default palette, recent-palettes list size, Lospec UI visibility.
Panels	Default visibility for Toolbox, Layers, Edit Bar, Advanced Bar, Preview, Character Map, Drawer, Color Mixer.
Audio	SFX and music enable/disable, master volume, mute.
Fonts	Default font and size, paths to scan for TTF/OTF.
Appearance	Various color scheme configurations.
Directories	Where DRAW looks for templates, palettes, music.

Enable AI Features

AI image generation is **off by default**, and while it is off DRAW surfaces nothing about it — no AI menu, no commands, no layer markers, no status text. Some people want nothing to do with generative tooling, and a disabled-but- visible feature is still in the way.

Tick **Enable AI Features** in the General tab to turn it on. That checkbox is the only place AI is mentioned while disabled. See Ch. 21 — AI Image Generation.

User fonts

Drop your own fonts into DRAW's **user data directory** and they show up in the font pickers — no reinstall, no config editing. The base folder is `<data>/FONTS/` (Linux `~/.local/share/DRAW/FONTS/`, with the platform data-dir equivalent on macOS/Windows). Where each type goes:

Font type	Extensions	Where to put it

TTF / OTF	`.ttf` `.otf`	`FONTS/` root **or** any subfolder
Bitmap	`.fxx` /F16/F14/F08, `.psf` , `.bdf` , `.pcf` , `.fon`	a **subfolder** of `FONTS/`
Color bitmap (CBF)	DPaint-style `.bmp` spritesheets	a **subfolder** of `FONTS/`
TheDraw	`.TDF`	`FONTS/THEDRAW/` — plain `.TDF` files, no `.TDX` index needed

A subfolder's name becomes its group label in the font dropdown. Each source has a checkbox in the **Fonts** settings tab — **User TTF/OTF Fonts**, **User Bitmap Fonts**, **User TDF Fonts** — mirrored by `FONTS_INCLUDE_USER` , `FONTS_INCLUDE_USER_BITMAP` , and `FONTS_INCLUDE_USER_TDF` in `DRAW.cfg` . Toggles take effect on the next launch.

`DRAW.cfg`

`DRAW.cfg` is a plain-text key/value file that lives next to the executable. You can hand-edit it any time. There are also **OS-specific** variants — `DRAW.linux.cfg` , `DRAW.macOS.cfg` , `DRAW.windows.cfg` — that override the base file when present.

CLI flags:

- `--config /path/to/your.cfg` — use a non-default config file.
- `--config-upgrade` — reconcile your existing config with any new defaults introduced by an upgrade. Recommended after each release.
- `--option KEY=VALUE` — override any config key from the command line (repeatable; beats the `.cfg`). E.g. `--option FONTS_INCLUDE_USER_TDF=TRUE` .
- `--options-list` — print every config key with its default and description, then exit.

Theming — Icons, Colors & Sounds

> 🎯 **Goal:** Customize DRAW's look and feel.

A **theme** is a folder under `ASSETS/THEMES/` that contains:

- A `THEME.CFG` file with all UI colors.
- A directory of icon PNGs (replaceable per theme — see `ASSETS/THEMES/DEFAULT/IMAGES/`).
- A `SOUNDS/` folder of WAV/OGG files (per-theme SFX).
- A `MUSIC/` folder of tracker tunes that play on startup or via the Audio menu.
- A `FONTS/` folder of bitmap and vector fonts.
- A `splash.png` for the launch animation.

Anything you can theme — UI palette, transform-overlay frame, smart-guide colors, layer panel highlights — lives in `THEME.CFG`. **No recompile is required**; DRAW reloads themes at runtime.

Display & Toolbar scale

- **Display Scale** — 1× through 8×. Suits HiDPI monitors.
- **Toolbar Scale** — 1× through 4×. Independent of display scale, so you can have small toolbar icons on a HiDPI display.

Panel Layout & Docking

> 🎯 **Goal:** Arrange panels for your workflow.

Dockable panels

Panel	Toggle
Toolbox	`Tab`
Layer Panel	`Ctrl+L`
Edit Bar	`F5`
Advanced Bar	`Shift+F5`
Character Map	`Ctrl+M`
Preview Window	`F4`
Drawer	(Organizer / View menu)
Color Mixer	View → Color Mixer

Each can be docked **left or right** by `Ctrl+Shift` +clicking on the panel itself.

UI master toggles

- `F11` — toggle **all** UI (canvas-only mode).
- `Ctrl+F11` — keep only the menu bar.
- `F10` — toggle the status bar.

DRAW also supports **auto-hide** while drawing: panels fade out so they don't obscure your work, then return when the cursor leaves the canvas.

Cursor system

The cursor system uses your OS-native cursor for UI hovers and a custom-painted cursor for tool-specific feedback (crosshair on dot, brush footprint on brush, etc.). This is automatic and themeable.

➡️ Next: Chapter 13 — Audio: Music & Sound Effects

Ch. 13 🔊 Audio: Music & Sound Effects

> **What you'll learn:** DRAW's per-theme sound effect bank and tracker-music player, plus how to swap, mute, and customize the audio experience — including MIDI playback and SF2 SoundFont configuration.

Audio System — SFX, Music & Customization

> 🎯 **Goal:** Configure the creative audio experience.

Sound effects (21 categorized slots)

DRAW emits short SFX for distinct UI actions — menus, tools, selection, fill, clipboard ops, layer ops, text entry, sliders, drag-and-drop, and more. The sounds live in the active theme's `SOUNDS/` folder as WAV or OGG files. Replace any file with one of the same name to retheme that action without touching code.

You can:

- Globally enable / disable SFX.
- Adjust master SFX volume.
- Mute (independent of disabling).

Background music

DRAW plays **tracker** modules and **MIDI** files natively:

Format	Description
`.mod`	Amiga ProTracker / FastTracker.
`.xm`	FastTracker II Extended Module.
`.it`	Impulse Tracker.
`.s3m`	Scream Tracker 3.
`.rad`	Reality Adlib Tracker.
`.mid` / `.midi`	Standard MIDI Format (SMF).
`.rmi`	RIFF MIDI (Windows MIDI container).

The **Audio** menu (rightmost in the menu bar) hosts:

- **Auto-shuffle** — when the current track ends, DRAW picks a random next track from the active music folder.
- **Next track** — `}`
- **Previous track** — `{`
- **Random track** — picks at any time. Use the **Random Track** submenu to restrict by format: `.MOD`, `.IT`, `.XM`, `.RAD`, `.MID`, or `.RMI`.
- **Volume up / down** — ±10% increments, independent of SFX volume.
- **NOW PLAYING** — read-only display of the current track name and format (e.g. `NOW: jester - stardust [MOD]`). Name is truncated to 32 characters.
- **Explore Music Folder** — opens the music directory in your OS file manager.

All audio settings persist in `DRAW.cfg`.

MIDI Playback & SoundFont Configuration

> 🎯 **Goal:** Get the best possible MIDI sound quality.

DRAW uses QB64-PE's built-in **OPL3 FM emulator** for MIDI/RMI playback out of the box — no extra files required. For richer, more realistic MIDI audio you can supply your own **SoundFont2** (`.sf2`) file.

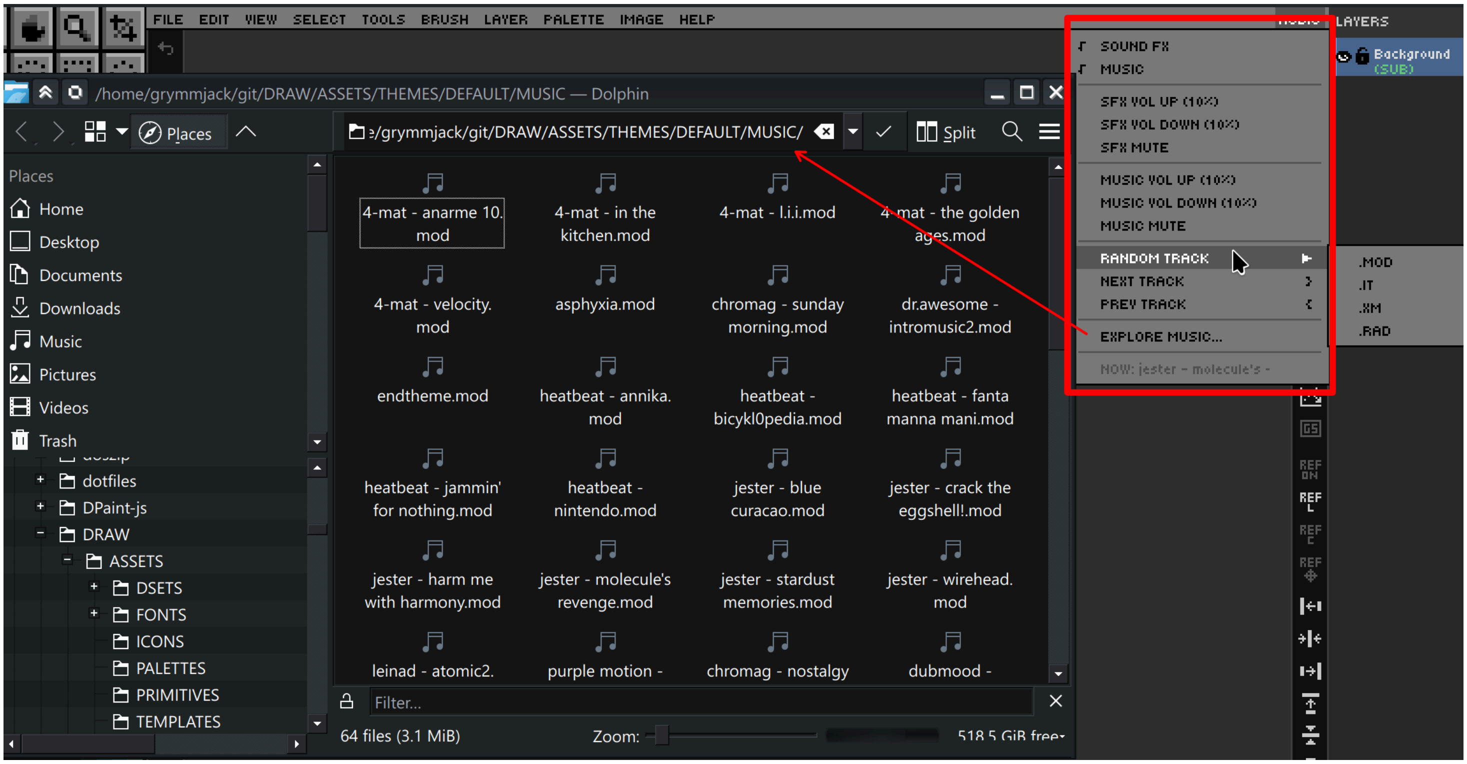
Configuring an SF2 SoundFont

1. Open **Settings** (Edit → Settings or `Ctrl+,`).
2. Navigate to the **Audio** tab.
3. Scroll to the **MIDI Soundfont** section.
4. Click the `...` browse button to locate a `.sf2` file on your system.
5. Click **Apply** or **OK** — the font is loaded before every MIDI/RMI track.
6. To revert to built-in OPL3 FM synthesis, click the **Clear** button next to the path field.

> **Tip:** Free general-MIDI soundfonts such as *GeneralUser GS* or *SGM-V2.01* work well and are widely available online.

How MIDI playback works

- `_MIDISOUNDBANK` is called with the SF2 path immediately before `_SNDOPEN` for each MIDI/RMI track.
- If the SF2 path is empty or the file no longer exists, DRAW automatically falls back to the built-in OPL3 emulator.
- The `MIDI_SF2_FILE` key in `DRAW.cfg` stores the path so it persists across sessions.
- Drop `.mid`, `.midi`, or `.rmi` files into `ASSETS/THEMES/DEFAULT/MUSIC/` and they will be included in random-track auto-shuffle alongside tracker modules.



➡️ Next: Chapter 14 — Pixel Art Analyzer

Ch. 14 🔍 Pixel Art Analyzer

> **What you'll learn:** How DRAW's built-in Pixel Art Analyzer detects the five most common pixel-art mistakes and how to fix each one.

Pixel Art Analyzer — Find & Fix Common Issues

> 🎯 **Goal:** Improve pixel art quality with automated analysis.

The **Pixel Art Analyzer** scans the active layer (or selection) for five well-known pixel-art problems and visualizes each one in an overlay so you can fix them efficiently.

Detection categories

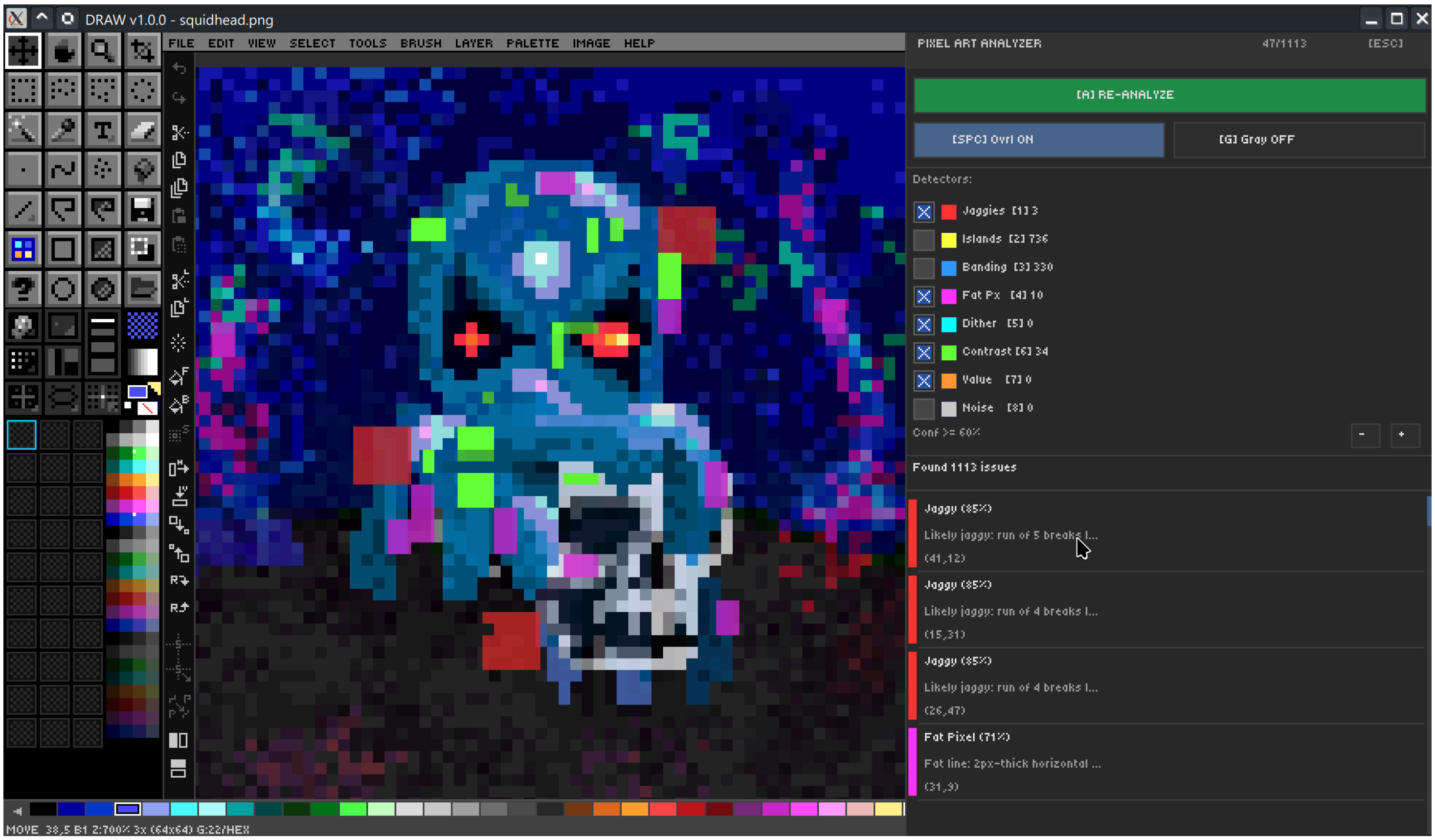
Category	What it finds	Why it's a problem
Orphan pixels	Single isolated pixels surrounded by transparency or different-colored pixels.	Visual noise and unintentional speckling.
Jagged lines	Stairstepped lines with uneven step lengths.	Reads as rough or amateurish; clean stairsteps should be regular.
Banding	Repeating horizontal/vertical color stripes that should be a smooth gradient.	Looks like a quantization artefact, not intentional shading.
Pillow shading	Concentric layered shading that follows the silhouette outline.	Produces a "puffed up" 3D effect that ignores light direction.
Doubles	Two adjacent identical-color pixels along an outline that should be 1px.	Throws off the line weight and stair-step rhythm.

How it works

The analyzer uses a precompute engine for fast incremental analysis. After the first scan it caches the per-pixel classifications so re-analysis after a fix is near-instant. Results are presented in an interactive dialog with totals per category and a clickable list that pans/zooms to each occurrence.

🎨 **Try it — before/after cleanup**

1. Open a sprite that you know is rough.
2. Run `Image → Pixel Art Analyzer`.
3. Step through orphan pixels first; remove or merge each one.
4. Re-analyze. Move to jagged lines next, then doubles, banding, and pillow shading last.
5. Save before-and-after copies for your portfolio.



➡️ Next: Chapter 15 — Reference Image & Import

Ch. 15 🖼️ Reference Image & Import

> **What you'll learn:** How to load a reference image for tracing or study, and how to import oversized images with interactive placement, rotation, and flipping.

Reference Image — Tracing Made Easy

> 🎯 **Goal:** Use reference images for tracing and study.

Press `Ctrl+R` to toggle a reference image. The reference renders **behind every layer** with adjustable opacity, so it never interferes with the artwork you're producing.

Action	Shortcut
Toggle reference image	`Ctrl+R`
Adjust reference opacity	`Ctrl+Shift` +Wheel (5–100%)
Reposition / zoom / nudge	Reference toolbar buttons

Reference image state is persisted in the `.draw` file along with its opacity, position, and zoom — so a project's tracing setup is never lost.

> 🎨 **Try it — pixel art from a photo**
>
> 1. Load a photo as a reference image.
> 2. Set opacity to 50%.
> 3. Pick a limited palette (e.g., Endesga 32).
> 4. Trace key shapes on a layer above with the line and polygon tools.
> 5. Fill regions with flood fill.
> 6. Hide the reference (`Ctrl+R`) to evaluate the standalone result.

Image Import — Oversized Placement & Transform

> 🎯 **Goal:** Import and position external images.

`File → Import Image` is different from `File → Open Image`. Open *replaces* your canvas; Import places the incoming image as a **floating overlay** that you position interactively before it commits as a layer.

While in import mode:

- Pan and zoom *inside* the imported image to crop tightly.
- Rotate 90° CW / CCW with the import toolbar buttons.
- Flip horizontal / vertical.
- Drag the destination box's resize handles to scale the placement.
- Live preview reflects every transform.

This is the cleanest way to bring an external sprite into a project at the right size and orientation, without permanent destructive resampling.

Aseprite & Photoshop

DRAW reads `.ase` / `.aseprite` and `.psd` files directly via `File → Open Aseprite` and `File → Open Photoshop`. Layers are preserved (where they map cleanly to DRAW's blend mode set) and re-exposed as native DRAW layers.

➡️ Next: Chapter 16 — Keyboard Shortcuts & Command Palette

Ch. 16 ⌨️ Keyboard Shortcuts & Command Palette

> **What you'll learn:** How to find any command instantly via the Command Palette, and the most important shortcuts grouped by activity.

Command Palette — 200+ Commands at Your Fingertips

> 🎯 **Goal:** Access any command instantly.

Press `?` to open the **Command Palette**. Start typing — DRAW does fuzzy-matching across command names so you can type `flip` to find both *Flip Horizontal* and *Flip Canvas Vertical*. Each match shows its hotkey on the right; arrow keys to highlight, Enter to run.

Use `Help → Cheat Sheet` for **Quick Reference Mode**, which lists every command in the application without filtering — handy for learning what's available.

The Palette is grouped into categories that match the menus:

- Tools
- File
- Edit
- View
- Color
- Brush
- Layer
- Canvas
- Assist
- Grid
- Symmetry
- Select
- Help
- Image
- Audio

Keyboard Shortcuts Cheat Sheet

> 🎯 **Goal:** Memorize the essential shortcuts.

The full, authoritative list is in `SHORTCUTS.md` at the repository root — generated directly from the input registry, so it never drifts from the app, and it covers **both keyboard and mouse**. Below are the essentials — what you'll use thousands of times a session.

Tool selection (single key)

Key	Tool
`B`	Brush
`D`	Dot
`L`	Line
`R`	Rectangle (`Shift+R` filled)
`C`	Ellipse (`Shift+C` filled)
`P`	Polygon (`Shift+P` filled)
`F`	Fill
`K`	Spray
`I`	Picker
`E`	Eraser
`M`	Marquee
`W`	Wand
`V`	Move
`Z`	Zoom
`T`	Text
`Q`	Bezier
(Toolbar)	Smart Shapes (Polygon, Pie/Donut, Rounded Rect, Tab, Pill, Pac-Man, 3D Dice, Bevel Rect, Arrow, 3D Text)

Essential combos

Combo	Action
`Ctrl+S` / `Ctrl+O` / `Ctrl+N`	Save / Open / New
`Ctrl+Z` / `Ctrl+Y`	Undo / Redo
`Ctrl+C` / `Ctrl+X` / `Ctrl+V`	Copy / Cut / Paste
`Ctrl+A` / `Ctrl+D` / `Ctrl+E`	Select All / Deselect / Clear
`Ctrl+L`	Toggle Layer Panel
`[` / `]`	Brush Size – / +
`\`	Brush Shape (circle ↔ square)
`X`	Swap FG / BG
`0` – `9`	Paint Opacity 100% / 10–90%
`'` `;` `.` `,`	Grid: toggle / snap / size+ / size−
`/`	Toggle grid alignment (Corner ↔ Center)
`G+Arrow`	Adjust grid width/height independently
`}` / `{`	Music: next / previous track
`F4` – `F9` , `F11`	Panel and mode toggles
`Tab`	Toggle Toolbar
`?`	Command Palette

Smart Shapes & 3D Tool Modifiers

Key	Function
`Up/Down/Left/Right`	Adjust shape parameters (segments, bevel, mouth, Z depth, etc.)
`Mouse Wheel`	Adjust primary parameter
`Shift`	Constrain aspect ratio
`Left-click drag`	Wireframe mode (FG color)
`Right-click drag`	Solid mode (BG color fill, FG color wireframe)
`4` / `6` / `8` / `0` / `1` / `2`	Switch dice type (D4/D6/D8/D10/D12/D20)
`W` / `A` / `S` / `D` / `Q` / `E`	Orbit light
`=` / `-`	Change light elevation
`L` + `0..9`	Light intensity presets

Bezier Tool

Key / Action	Function
`Q`	Activate Bezier tool
Click	Drop corner anchor
Click+drag	Drop smooth anchor, shape handle
`H`	Toggle handle visualisation
`Backspace`	Remove last anchor
`Enter`	Commit curve
`Escape`	Cancel

Line Tool — End Caps

Key	Function
`s`	Cycle start cap (while dragging)
`e`	Cycle end cap (while dragging)

Hold-Key Combos (Non-Modifier)

The following combos require holding a non-modifier key (not Ctrl/Alt/Shift) and clicking, pressing, or dragging another input. These are distinct from standard modifier shortcuts and are easy to miss!

Hold Key	+ Action	Effect
`F`	Left Click	Global Fill — flood-fills all visible non-locked layers with FG color (contiguous)
`F`	Right Click	Global Fill — flood-fills all visible non-locked layers with BG color (contiguous)
`Shift+F`	Left Click	Replace-all Global Fill — replaces all matching pixels across all visible layers with FG color
`Shift+F`	Right Click	Replace-all Global Fill — replaces all matching pixels across all visible layers with BG color
`E`	Click	Flood Erase — erases contiguous same-color pixels to transparent (no selection step)
`W`	Click	Magic Wand — select from merged canvas (all visible layers)
`G`	Arrow Key	Resize grid width/height independently
`L`	Digit (0-9)	Set Smart Shape 3D light intensity preset
`R`	Click	Picker — sample from reference image instead of visible layers

> See also: Tool-specific sections for additional drag-time and context-sensitive combos (e.g., Smart Shapes, Line End Caps, etc.).

For everything else, use the Command Palette or read the chapter where the feature is introduced.

Customize Controls — Rebind Anything (New in 2.1.0)

> 🎯 **Goal:** Make DRAW's controls match your muscle memory.

Every shortcut in DRAW is now **rebindable** — keyboard *and* mouse. Open **Edit** → **Customize Controls** (or search "Customize Controls" in the Command Palette).

Rebinding a control

1. Type in the **FIND** box to filter the list by command name.
2. Click **SET...** on a row to open the capture dialog.
3. Press the new key, click a **mouse button**, spin the **wheel**, or **tilt** the wheel left/right — whatever you press is captured. Toggle the Ctrl/Shift/Alt checkboxes to require modifiers.
4. If the input is already bound elsewhere, DRAW **warns and steals** it (the old binding is unbound). Click **OK** to apply — changes take effect immediately.

Rows you've changed from the factory default show a marker and a "= changed from default" legend. **RESET ALL** restores every control to the shipped defaults instantly.

Mouse controls you can bind

Control	Default	Notes
Vertical wheel	Zoom in/out (cursor-centered)	Rebindable; can be set to canvas pan or brush size
Wheel **tilt** (left/right)	Brush size − / +	Tilt right = bigger; rebind to **Pan Left/Right** for horizontal scroll
`Ctrl` + wheel	Brush size	
Thumb buttons (Back / Forward)	Undo / Redo	
Right-click (Zoom tool)	Zoom **out**	Left-click zooms in
Pan drag button	Middle / Space+drag	Rebindable

There are also dedicated **Pan Up / Down / Left / Right** actions you can bind to the wheel or tilt for scroll-to-pan.

Presets & export

- **Load preset** applies a full keymap in one click: *Aseprite, Photoshop, GIMP, Krita, Illustrator, Procreate, MS Paint, Pro Motion, DeluxePaint.*
- **Export as...** writes your live keymap (including overrides) to **Markdown, HTML, or PDF**.
- Bindings persist across sessions in `DRAW.bindings`; the CLI flag `--load-preset <name>` applies one at launch.

➡️ Next: Chapter 17 — Undo, Redo & History

Ch. 17 ↩️ Undo, Redo & History

> **What you'll learn:** How DRAW's unified history system works, why text-local undo is separate, and why you can experiment fearlessly.

History System — Fearless Experimentation

> 🎯 **Goal:** Understand the unified undo/redo system.

DRAW has **one timeline** for everything you do — drawing, transforms, layer ops, palette ops, even the import overlay. Press `Ctrl+Z` to step backward and `Ctrl+Y` to step forward.

How history is recorded

History records **on mouse release**, automatically. There is no manual save-state command, and there is a guard against double-recording on the same frame. Each entry carries a human-readable label — *"Brush stroke on Layer 3"*, *"Move 12px on selected layers"*, *"Apply Transform"*, for a future history panel and comments in QB64PE Project Export.

Multi-layer awareness

When an operation affects multiple layers (a multi-layer select-and-flip, a Smart Erase across visible layers, a group merge, etc.), DRAW records a single composite history step. Smart Erase additionally tracks per-layer erasure so undo / redo behaves intuitively per layer.

Text-local undo

The text tool has its own 128-state local history while you are editing a text layer. `Ctrl+Z` and `Ctrl+Y` operate on that local history during text entry — typo experiments don't pollute the global timeline. As soon as you commit (`Esc`), the entire edit becomes a single global history entry.

Try things

The single biggest productivity unlock in DRAW is: **try things**. Symmetry, palette ops, blend modes, custom-brush fills, and transform overlays are all undoable. There is no "save first" anxiety — `Ctrl+Z` always works.

> 💡 **Tip:** When you make a destructive-looking change (palette ops, image adjustments), don't try to remember the parameters. Just `Ctrl+Z` and `Ctrl+Y` to flip-flop and visually compare.

➡️ Next: Chapter 18 — Real-World Pixel Art Workflows

Ch. 18 🎓 Real-World Pixel Art Workflows

> **What you'll learn:** Step-by-step recipes for the most common pixel-art tasks: a 16×16 sprite, a seamless tile, isometric building, mandala, ANSI art, sprite-sheet assembly, and converting a photo into pixel art.

Workflow — Game Sprite (16×16 Character)

> 🎯 **Goal:** Create a complete game character sprite.

1. `Ctrl+N` — new canvas, 16×16, grid = 1, snap on.
2. **Choose a limited palette** — NES or PICO-8 are great starting points.
3. **Layer 1: Silhouette** — draw the outline in a single dark color.
4. **Layer 2: Base colors** — turn on Opacity Lock and fill regions.
5. **Layer 3: Shading** — set blend mode to Multiply, paint cool desaturated tones in shadow areas.
6. **Layer 4: Highlights** — set blend mode to Screen, dab warm light tones on edges facing your imaginary light source.
7. **Run the Pixel Art Analyzer** — fix orphans, doubles, jaggies, banding.
8. **Export** — File → Export As → PNG (plain) for engines, or save as `.draw` to keep the layers editable.

Workflow — Seamless Tile Texture

> 🎯 **Goal:** Create a tileable background pattern.

1. `Ctrl+N` — new canvas, 32×32 or 64×64.
2. **View → Pattern Tile Mode** — enable Pattern Tile Mode. The 3×3 preview will reveal seams instantly.
3. Draw your base pattern.
4. Watch the seams in the preview. Fix them at the canvas edges by painting through the wrap.
5. Capture the result as a custom brush.
6. On a fresh test canvas, flood-fill with the brush to verify seamlessness.
7. Export as a plain PNG tile.

Workflow — Isometric Pixel Art

🎯 **Goal:** Create isometric buildings and objects.

1. `Ctrl+'` — set grid geometry to Isometric.
2. `;` — enable snap.
3. The grid guides ensure perfect 2:1 ratio. (Also try Edit → Angle Snap: Pixel Art)
4. Use the Polygon tool for angled surfaces (top, left, right faces of a cube).
5. Put each face on its own layer.
6. Use Multiply / Screen blend modes for light/shadow on each face.
7. Group the layers per object so you can move it as a unit.

Workflow — Symmetrical Mandala / Pattern

> 🎯 **Goal:** Create complex symmetrical artwork.

1. `F7` **× 3** — enable 8-way (asterisk) symmetry.
2. `Ctrl+Click` the canvas center.
3. Draw one slice — eight reflections appear instantly.
4. Layer subsequent strokes on different layers with different blend modes.
5. Vary opacity for depth.
6. Use custom-brush stamps (small flowers, dots, geometric shapes) for ornamentation.

Workflow — ANSI / Text Art with Character Mode

🎯 **Goal:** Create text-mode art using block characters.

1. Press `T`, switch to the VGA font.
2. Enable Character Mode.
3. Open the Character Map (`Ctrl+M`).
4. Use `F1`–`F12` for the ANSI block-shading characters (`░ ▒ ▓ █ ▀ ▄ ▌ ▐`).
5. Navigate with the virtual cursor.
6. `Alt+U` picks colors from any existing character cell.
7. The DOT and RECT tools fill cells with the active glyph instead of pixels.
8. Toggle CP437 mode (View menu → Charmap Unicode/CP437 Toggle) for the classic DOS feel.

Workflow — Sprite Sheet Assembly & Export

> 🎯 **Goal:** Assemble sprites into a sheet and re-extract.

1. Create individual sprites on separate layers.
2. Use grid + snap for consistent spacing.
3. Use **layer groups** per animation frame.
4. Use **Align & Distribute** for perfect layout.
5. Export the full sheet as PNG.
6. Use **Extract Images** to decompose the sheet into individual PNGs (Chapter 10).
7. Save the project as `.draw` to preserve layers for future edits.

Workflow — Photo to Pixel Art

> 🎯 **Goal:** Convert a photograph into pixel art.

1. `Ctrl+R` — load the photo as a reference image.
2. Choose a limited palette (Endesga 32 is a great default).
3. Set reference opacity to ~60% with `Ctrl+Shift` +Wheel.
4. Trace key shapes on a new layer with line and polygon tools.
5. Fill regions with flood fill.
6. Add detail with the brush and dot tools.
7. Apply Image Adjustments (Levels, Hue/Sat) for color tweaking.
8. Run **Posterize** if you want pixel-art-friendly value steps.
9. Hide the reference and evaluate the standalone result.

➡️ Next: Chapter 19 — Tips, Tricks & Advanced Techniques

Ch. 19 💡 Tips, Tricks & Advanced Techniques

> **What you'll learn:** Ten time-saving tips that experienced DRAW users rely on every day, plus advanced layer techniques for non-destructive workflows.

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10 Time-Saving Tips You Need to Know

> 🎯 **Goal:** Boost productivity with pro tips.

1. `Alt` **+Click is a temporary picker — *with any tool that can drop pixels.*** No need to switch to the Picker tool for a quick eyedrop.
2. **Middle-click drags pan — *with any tool.*** Stop reaching for the Pan tool just to nudge the view.
3. **Hold `E` for a temporary eraser.** Release `E` and you're back to the previous tool. Lifesaving for cleanup mid-stroke.
4. `Shift` **+Right-click draws a connecting line** from the previous click to the current cursor — works in Brush, Dot, Line.
5. `?` **(Command Palette) is faster than menu diving.** If it takes more than two clicks to find a command, you're doing it wrong.
6. **Opacity Lock + a giant brush = instant recolor.** No selection needed. Just lock the layer and slosh.
7. **Custom brush + tiled fill = instant pattern fills.** Capture a small motif as a brush, then flood fill an area.
8. `Ctrl+Shift+C` **is Copy Merged.** Flatten the visible composite to clipboard without merging your actual layers.
9. **Double-middle-click = instant pan/zoom reset.** Get back to a fitting view in one gesture.
10. `.draw` **is a PNG you can open in any viewer.** Show off your art on Discord without exporting first — your project file *is* the preview.
11. **Move shapes while dragging:** Hold SPACE while dragging any geometric or Smart Shape tool (Line, Rectangle, Ellipse, Polygon, Smart Shapes) to temporarily reposition the in-progress shape with the mouse, instead of resizing it. Release SPACE to resume resizing from the new anchor. Grid snap is honored. (Photoshop-style move-during-drag.)

Advanced Layer Techniques

> 🎯 **Goal:** Push the layer system to its limits.

Non-destructive color adjustment workflow

Build a stack with the artwork at the bottom, a Multiply layer for shadows above it, a Screen layer for highlights, and a Color-mode layer at the top for the overall colorway. Tweak the upper layers without ever touching the base. When the look is right, optionally merge down — but consider keeping the layers for easy revisions.

Group opacity for complex transparency

A group's opacity slider acts on the entire group as if it were a single rasterized layer. This is how you fade an entire compound element (e.g., a complete UI panel made of many layers) without losing the per-element relationships.

Pass Through mode

A group set to **Pass Through** lets each child blend with what's *below the group*, not just within. This is essential for blend-heavy layers (Multiply / Screen / Add) inside a group, where Normal-mode group compositing would otherwise isolate them.

`Alt` +Drag with the Move tool — clone stamping

Holding `Alt` while dragging with the Move tool clones the dragged content into a new layer at the destination. Your original stays put.

Solo mode

`Alt` +Click any layer's eye icon to **solo** it — every other layer is hidden. Click again to restore. Indispensable for diagnosing a stack you've lost track of. Also handy is using the Move tool, and `Shift+Click` to select the layer under the pointer.

Multi-layer selection for bulk operations

Select multiple layers (`Ctrl` +Click / `Shift` +Click) and run any of: clear, fill, flip, rotate, scale, transform overlay, or alignment. The history records a single composite step.

Layer alignment for sprite-sheet layout

Align Left / Center / Right and Distribute Horizontally are the fastest way to organize a row of sprites — far faster than nudging by hand even with snap.

➡️ Next: Chapter 20 — Appendix: Quick Reference

Ch. 20 📋 Appendix: Quick Reference

> **What you'll learn:** Where the canonical cheat sheet lives, plus a one-page summary of the most-used keyboard shortcuts, the 56 bundled palettes, and all 19 blend modes.

Canonical reference

The complete, always-up-to-date list of keyboard **and mouse** shortcuts lives in `SHORTCUTS.md` at the repository root — generated directly from the input registry. The summary below is curated for quick visual scanning — when in doubt, the canonical file wins. All of these are rebindable via **Edit → Customize Controls** (see Chapter 16).

Keyboard shortcut summary

Tool keys

Key	Tool	Key	Tool
`B`	Brush	`M`	Marquee
`D`	Dot	`W`	Magic Wand
`L`	Line	`V`	Move
`R`	Rect (`Shift+R` filled)	`Z`	Zoom
`C`	Ellipse (`Shift+C` filled)	`T`	Text
`P`	Polygon (`Shift+P` filled)	`I`	Picker
`F`	Fill	`E`	Eraser
`K`	Spray		

Modifier vocabulary

Modifier	Default meaning
`Shift`	Constrain / add to selection / draw from center / fast-scroll / Smart Erase.
`Ctrl`	Perfect aspect (square / circle) / command keystrokes.
`Alt`	Subtract from selection / temporary picker / clone (with Move).
`Ctrl+Shift`	Angle snap / bypass grid snap.

Panel toggles

Key	Panel
`Tab`	Toolbar
`F4`	Preview Window
`F5`	Edit Bar
`Shift+F5`	Advanced Bar
`Ctrl+L`	Layer Panel
`Ctrl+M`	Character Map
`F10`	Status Bar
`F11`	All UI
`Ctrl+F11`	Menu Bar only

Canvas / view

Action	Shortcut
Zoom in / out	`Ctrl+=` / `Ctrl+-`, Mouse Wheel (Zoom tool: right-click zooms out)
Brush size – / +	`[` / `]`, `Ctrl`+Wheel, wheel tilt left/right
Reset zoom	`Ctrl+0`
Reset pan + zoom	Double-middle-click
Pan	Middle-drag, `Space`+drag (or bind the wheel/tilt to Pan in Customize Controls)
Pattern tile	View menu / Advanced Bar
Grayscale preview	`Ctrl+Alt+Shift+G`
Reference image	`Ctrl+R`
Canvas border	`#`

Color

Action	Shortcut
Swap FG / BG	`X`
Reset FG / BG	*Palette menu → Default Colors*
BG = transparent	`Shift+Delete`
Paint opacity	`0`–`9`
Temp picker FG / BG	`Alt`+L-click / R-click

Brush

Action	Shortcut
Size – / +	`[` / `]`
Shape	`\`
Preview footprint	hold `` ` ``
Pixel Perfect	`F6`
Recolor mode	`F9`
Outline mode	`Shift+O`

Selection / clipboard

Action	Shortcut
Select All / Deselect	`Ctrl+A` / `Ctrl+D`
Invert	`Ctrl+Shift+I`
Copy / Cut / Paste	`Ctrl+C` / `X` / `V`
Paste in Place	*Edit menu*
Copy Merged	`Ctrl+Shift+C`
Copy / Cut to New Layer	`Ctrl+Alt+C` / `X`

Layers

Action	Shortcut
New / Duplicate / Delete	`Ctrl+Shift+N` / `D` / `Delete`
Group / Ungroup	`Ctrl+G` / `Ctrl+Shift+U`
Group selected	`Ctrl+Shift+G`
Move up / down	`Ctrl+PgUp` / `Ctrl+PgDn`
Move to top / bottom	`Ctrl+Shift+]` / `Ctrl+Shift+[`
Merge Down / Visible	`Ctrl+Alt+E` / `Ctrl+Alt+Shift+E`

History / Edit

Action	Shortcut
Undo / Redo	`Ctrl+Z` / `Ctrl+Y`
Clear (selection or layer)	`Delete` or `Ctrl+E`
Fill FG / BG	`Backspace` / `Shift+Backspace`

Grid / Symmetry

Action	Shortcut
Toggle grid	`'`
Pixel grid	`Shift+'`
Snap to grid	`;`
Grid size − / +	`,` / `.`
Cycle geometry	`Ctrl+'`
Cycle symmetry	`F7`

AI (only when AI features are enabled)

Action	Shortcut
Cancel AI generation / batch	`Ctrl+Alt+K`

See Ch. 21 — AI Image Generation.

All 56 bundled palettes

Located in `ASSETS/PALETTES/`. The full set includes:

- **Console classics** — NES, PICO-8, Commodore 64, Game Boy (DMG and Pocket), MSX, Amiga, Atari.
- **Arcade / DOS** — CGA, EGA, VGA.
- **Modern curated** — Endesga 32 / 64, DawnBringer 16 / 32, AAP-64, Sweetie 16, Resurrect 64.
- **Lospec greatest hits** — and 40+ more available via DRAW's Lospec integration.

All 19 blend modes

Family	Modes
Basic	Normal · Multiply · Screen · Overlay
Math	Add (Linear Dodge) · Subtract · Difference
Comparison	Darken · Lighten
Dodge / Burn	Color Dodge · Color Burn
Light	Hard Light · Soft Light · Vivid Light · Linear Light · Pin Light
Color	Exclusion · Color · Luminosity
Group only	Pass Through

Chapter 21 — 🤖 AI Image Generation

> 🎯 **Goal:** Generate pixel art from a text prompt using an external generator of your choosing, and work with the results as ordinary layers.

Off by default

AI features are **disabled until you ask for them**, and while disabled DRAW shows nothing about them at all — no AI menu, no commands, no layer markers, no status text. Enable them in **Settings → General → Enable AI Features**.

Turning the feature off again hides it completely, including on projects that already contain AI layers: those layers keep working and simply render as ordinary layers.

DRAW does not talk to a model

DRAW runs an **external command-line generator that you configure**. It builds a command, runs it, and imports the PNG that comes back. You choose the engine, the models, and whether anything leaves your machine.

The bundled example is pixelmon, a local ComfyUI + SDXL pipeline. A test engine (`DEV/ai-echo.sh`) is also included so you can exercise the whole pipeline in about a second.

Generators live in `DRAW.ai.cfg` in your config directory. **AI → Tools…** edits them; the first time you use an AI command with nothing configured, DRAW offers to write a starter file.

Generating

AI → New AI Layer… opens the generate dialog.

Field	Notes
Tool	Which configured generator to run
Style	Optional style guide — see below
Preset	A saved prompt; picking one loads it into the prompt box
Prompt	Multi-line, word-wrapped, with selection and clipboard
Size	`[IMAGE SIZE]` fills in the canvas size; `[IMAGE xN]` appears only when your display is scaled
Position	3×3 preset grid, or type X/Y directly
Seed	Same seed reproduces a result; `[RANDOM]` re-rolls
Count	Above 1 generates a batch — see below
SURPRISE ME	Random style, random saved prompt, fresh seed

Generation runs **in the background**. DRAW stays fully usable, and the status bar shows an animated progress line with the elapsed time. Press `Ctrl+Alt+K` to abort — this terminates the generator process, it does not merely stop watching it.

Working with a selection

With a selection active, the dialog starts with the selection's **size** and its **top-left corner**. The result is scaled to fit inside the selection with its aspect ratio preserved, so nothing is skewed: the limiting dimension fills the selection exactly and the other is centred.

Enlarging is restricted to **whole-number factors**, because scaling pixel art by a fractional amount resamples it into mush. A small result inside a large selection therefore sits centred at 1×, 2× or 3× rather than being smeared to fill.

Batch generation

Set **Count** above 1 (up to 20) to generate several variations at once. They land in a **layer group**, and every item shares identical settings — same tool, style, prompt, size and position — differing **only by seed**. A batch is a set of variations on one idea, not a set of unrelated images.

Seeds run from the seed shown in the dialog upward, so re-running a batch with the same starting seed reproduces the same set.

Items are generated one at a time. The status bar shows progress as `[2/10]`, and `Ctrl+Alt+K` stops the remainder — anything already generated is kept.

AI layers

A generated layer is marked `[AI]` in the layer panel. Hovering it shows the full prompt, with the engine, model, style and seed beneath.

Right-click an AI layer for:

- **Edit Prompt...** — change the stored prompt without regenerating
- **Regenerate** — re-run it, pre-filled with the prompt and seed it was made with

The prompt, style, tool, seed and size are **saved in the `.draw` file**, so a layer can be regenerated — or reproduced exactly — after reloading a project.

Styles and prompts

AI → Style Editor... manages style guides. A style can:

- wrap your prompt with a **prefix** and **suffix**
- add its own **CLI arguments** (e.g. pixelmon's `--style ega`)
- be **scoped to one tool**, since a style name means nothing to a different generator
- **lock a seed**, so its look is reproducible

> ⚠️ Command-line flags belong in a style's **Args** field, not its prefix or suffix. Prefix and suffix are *prompt text* — a flag placed there is passed inside the quoted prompt and the generator never sees it as a flag. DRAW migrates obviously-misplaced flags automatically and notes it in the log.

AI → Prompt Editor… manages saved prompt presets — plain text starting points you can pick from the generate dialog.

Argument macros

A tool's arguments are a **template**. DRAW expands `{macros}` before running it, so one configuration adapts to whatever you are working on.

Click **[?]** in the tool or style editor for a live reference listing every macro alongside **its current value**, so you can see exactly what your tool will receive.

Group	Macros
Run	`{prompt}` `{style}` `{seed}` `{outdir}` `{outname}`
Document	`{if}` `{id}` `{iw}` `{ih}`
Selection	`{sw}` `{sh}`
Layer	`{numlayers}` `{lidx}` `{lname}` `{lw}` `{lh}` `{lx}` `{ly}`
Brush / grid	`{bw}` `{bh}` `{gw}` `{gh}`
Colour	`{pal}` `{numpal}` `{fg}` `{bg}` `{pmode}`
Context	`{tool}` `{px}` `{py}` `{font}` `{fsize}` `{fontfile}`
Steering	`{limg}` `{dimg}` `{bimg}`

Three rules apply to every macro:

1. **Values are whitespace-trimmed.**
2. **An UPPERCASE name yields an UPPERCASE value** — `{PAL}` gives `ANSI32`, `{pal}` gives it as-is. `{pal:lower}` and `{pal:slug}` are also available.
3. `sc*` **variants give the screen-scaled value** — `{sciw}` is the image width multiplied by the display scale. Every dimension and position macro has one.

DRAW supplies `{outdir}` as a fresh directory per run and imports the newest PNG it finds there, so tools that name their own output files work unchanged.

Steering images

`{limg}`, `{dimg}` and `{bimg}` export the **current layer**, the **flattened document** and the **custom brush** as PNGs, for generators that accept a reference image. `{dimg}` is the artwork only — never DRAW's interface.

The log

Every run is recorded in `ai.log` in your config directory: the full command line, the exit code, the tool's own output, and the imported file or the reason it failed. When a generation misbehaves, that file will tell you why. A failure dialog offers to open it for you.

Exercises

1. Enable AI features and let DRAW write the starter config. Generate once with the bundled `echo (test)` engine to confirm the pipeline end to end.
2. Make a selection, then generate into it. Note that the dialog pre-fills the selection's size and corner.
3. Create a style with a suffix of `pixel art, bold outlines` and generate the same prompt with and without it.
4. Run a batch of 4 and compare the variations — same prompt, four seeds.
5. Open the **[?]** macro reference and find three macros whose current values change when you switch tool or palette.